

NEP 2020

B.Sc. (Four Years) Computer Science Curriculum

with

Syllabus

**Department of Computer Science
Institute of Science**

Banaras Hindu University

B.Sc. (four Years) Computer Science Course Curriculum as per NEP-2020

I Semester

Code	Title	T	P	C	Course Category
CSCMJ101	Introduction to Programming using C	3	1	4	Major/Minor
CSCSE101	Web Designing* (Maximum Intake 80)	2	1	3	SEC-1
	From outside the Department	-	-	4	Minor
	From outside the Department	-	-	3	MDC-1
	From outside the Department	-	-	4	VAC-1
	From outside the Department	-	-	2	AEC-1
Total Credits				20	
CSCMD101	Fundamentals of Computing (Max Intake 80)	3	0	3	MDC-1

* Pre-requisite: Computer Science/Information Technology/Mathematics at 10+2 level

II Semester

Code	Course Title	T	P	C	Course Category
CSCMJ201	Data Structures and Algorithms	3	1	4	Major/Minor
CSCSE201	Mobile Application Development*(Maximum Intake 80)	2	1	3	SEC-2
	From outside the Department	-	-	4	Minor
	From outside the Department	-	-	3	MDC-2
	From outside the Department	-	-	2	VAC-2
	From outside the Department	-	-	2	VAC-3
	From outside the Department	-	-	2	AEC-2
Total Credits				20	
CSCMD201	Introduction to Information Technology (Max. Intake 80)	3	0	3	MDC-2

* Pre-requisite: Computer Science/Information Technology/Mathematics at 10+2 level

III Semester

Code	Course Title	T	P	C	Course Category
CSCMJ301	Digital Logic and Circuit Design	3	1	4	Major
CSCMJ302	Object Oriented Programming	3	1	4	Major
CSCSE301	Python Programming* (Maximum Intake 80)	2	1	3	SEC-3
	From outside the Department	-	-	2	AEC-3
	From outside the Department	-	-	4	Minor (VOC-1)
	From outside the Department	-	-	3	MDC-3
Total Credits				20	
CSCMV301	IT Tools and Applications(Max. Intake 80)	3	1	4	Minor (VOC-1)
CSCMD301	Introduction to Cyber Security (Max. Intake 80)	3	0	3	MDC-3

* Pre-requisite: Computer Science/Information Technology/Mathematics at 10+2 level

IV Semester

Code	Course Title	T	P	C	Course Category
CSCMJ401	Mathematics for Computer Science	3	0	3	Major
CSCMJ402	Database Management System	3	1	4	Major
CSCMJ403	Computer Organization	3	0	3	Major
CSCMJ404	Numerical Computing	3	1	4	Major/Minor
	From other Departments	-	-	4	Minor
	From other Departments	-	-	2	AEC-4
Total Credits				20	

V Semester

Code	Course Title	T	P	C	Course Category
CSCMJ501	Operating System Concepts	3	1	4	Major
CSCMJ502	Design and Analysis of Algorithms	3	1	4	Major
CSCMJ503	Theory of Computation	4	0	4	Major
CSCMJ504	Swayam Course <i>The course must be selected with dept approval.</i>	-	-	2	Major
CSCIN501	Internship	-	-	2	Internship
	From other Departments	-	-	4	Minor(VOC-2)
Total Credits				20	
CSCMV501	Introduction to Multimedia (Max. Intake 80)	3	1	4	Minor(VOC-2)

VI Semester

Code	Course Title	T	P	C	Course Category
CSCMJ601	Software Engineering	4	0	4	Major
CSCMJ602	Compiler Design	3	1	4	Major
CSCMJ603	Computer Networks	3	1	4	Major
CSCMJ604	Artificial Intelligence	3	1	4	Major
	From other Departments	-	-	4	Minor(VOC-3)
Total Credits				20	
CSCMV601	Data Analysis and Visualization (Max. Intake 80)	3	1	4	Minor(VOC-3)

VII Semester (Honours/Research)

Code	Course Title	T	P	C	Course Category
CSCMJ701	Machine Learning	3	1	4	Major/Minor
CSCMJ702	Research Methodology	2	0	2	Major
Electives (Any Two)					
CSCMJ703	Operations Research	3	1	8	Major
CSCMJ704	Cloud Computing	3	1		
CSCMJ705	Soft Computing	3	1		
CSCMJ706	Network Security and Cryptography	3	1		
CSCMJ707	Heuristics & Metaheuristics	3	1		
CSCMJ708	Knowledge & Data Discovery	3	1		
CSCMJ709	Distributed Systems	3	1		
CSCMJ710	Parallel Computing	3	1		
CSCMJ711	Image Processing	3	1		
CSCMJ712	Swayam Course <i>The course must be selected with dept approval.</i>	-	-	2	Major
	From other departments	-	-	4	Minor
Total Credits				20	

VIII Semester

Code	Course Title	T	P	C	Course Category
CSCMJ801	Deep Learning	3	1	4	Major/Minor
Electives (Any two)					
CSCMJ802	Internet of Things	3	1	8	Major
CSCMJ803	Blockchain Technology	3	1		
CSCMJ804	Information Retrieval	3	1		
CSCMJ805	Statistical Pattern Recognition	3	1		
CSCMJ806	Image Analysis and Computer Vision	3	1		
CSCMJ807	Natural Language Processing	3	1		
CSCMJ808	Data Compression	3	1		
CSCMJ809	Mobile Adhoc Networks	3	1		
CSCMJ810	Big Data Analytics	3	1		
CSCMJ811	Quantum Computing	3	1		
CSCMJ812	Swayam Course <i>The course must be selected with dept approval.</i>	-	-	4	Major
	From other departments	-	-	4	Minor
Total Credits				20	

VIII Semester (Research)

Code	Course Title	T	P	C	Course Category
CSCMJ801	Deep Learning	3	1	4	Major/Minor
CSCDS801	Project/Dissertation	-	-	12	
	From other departments	-	-	4	Minor
Total Credits				20	
The students from Computer Science Major may have choice to choose any one of the following in place CSCMJ801					
CSCMJ802	Internet of Things	3	1	4	Major
CSCMJ803	Blockchain Technology	3	1		
CSCMJ804	Information Retrieval	3	1		
CSCMJ805	Statistical Pattern Recognition	3	1		
CSCMJ806	Image Analysis and Computer Vision	3	1		
CSCMJ807	Natural Language Processing	3	1		
CSCMJ808	Data Compression	3	1		
CSCMJ809	Mobile Adhoc Networks	3	1		
CSCMJ810	Big Data Analytics	3	1		
CSCMJ811	Quantum Computing	3	1		

T→stands for number of lecture hours per week

P→stands for number of Practical hours per week

Semester-I

CSCMJ101	Introduction to Programming using C	Theory	Practical	Cumulative
		3	1	4
Major/Minor		45	30	75
Course Objectives				
• The course aims to provide exposure to problem-solving through programming. • It aims to train the student to the basic concepts of the C-programming language. • Students will learn to develop simple algorithms and flow charts to solve a problem. • Students will be able to develop logics which will help them to create programs, applications in C. • By learning the basic programming constructs they can easily switch over to any other language in future. • This course involves a lab component which is designed to give the student hands-on experience with the concepts.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Introduction to Programming, Program Concept, Characteristics of Programming, Stages in Program Development, Algorithms, Flowcharts, Types of Programming Methodologies, Introduction to C Programming Basic Program Structure in C, Variables and Assignments, Input and Output, Selection and Repetition Statements.			15
II	User defined functions, Standard library functions, Passing values between functions, Calling convention: Call by value and Call by reference, Recursive functions. Introduction to Arrays, Declaration and Referring Arrays, Arrays in Memory, Initializing Arrays. Arrays in Functions, Multi-Dimensional Arrays.			10
III	Introduction to String, Declaration and Initialization, Reading and Writing Strings, Arrays of Strings, String and Function, Standard String Library Functions. Basics of Pointers, Pointers and One-dimensional Arrays, Pointer Arithmetic, Pointers as Function Arguments, Pointers and Strings, Pointers and two-dimensional arrays, Arrays of Pointers, Storage Classes.			10
IV	Basics of Structures, Structures and Functions, Arrays of Structures, Pointers to Structures, Unions, Preprocessor, File Inclusion, Macro, Conditional Compilation, Dynamic Memory Allocation, Command Line Arguments, File Handling.			10
Texts / References Books				
1. Kernighan, B.W. and Ritchie, D.M., The C Programming Language, PHI. 2. Kanetkar, Y., Let Us C, BPB Publications. 3. Balagurusamy,E., Programming in ANSI C, McGraw-Hill.				
Learning Outcomes				
After the completion of this course the students will be able to:				
• Apply problem-solving knowledge and skills to write small, well-documented, effective C programs. • Choose the right data representation formats based on the requirements of the problem. • Able to write simple programs in the corresponding programming language to solve a task, given the constraints on the inputs. • Able to manually trace through a program to identify logical errors; • Able to differentiate between logical errors, syntax errors, and run-time errors. • Able to write code following good programming style (clear comments, naming convention, indentation, etc.)				

CSCSE101	Web Designing	Theory	Practical	Cumulative
		2	1	3
SEC		30	30	60
Course Objectives				
- Learn to create visually appealing and user-friendly websites using HTML, CSS, and JavaScript, focusing on design principles, layout techniques, and responsive design. - Develop skills in integrating advanced features and frameworks, such as interactive elements and backend integration, to build functional and dynamic web applications.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	HTML: Overview, Basic Tags, Elements, Attributes, Formatting, Phrase Tags, Meta Tags, Comments, Images, Tables, Lists, Text Links, Image Links, Email Links, Frames, Iframes, Blocks, Backgrounds, Colors, Fonts, Forms, Embed Multimedia, Marquees, Style Sheet, Layouts.			8
II	CSS: Overview, Syntax, Inclusion, Measurement Units, Colors, Background, Fonts, Text, Images, Links, Tables, Borders, Margins, Lists, Paddings, Cursors, Outlines, Dimension, Scrollbars, Visibility, Positioning, Layers, Filters, Media, Printing, Layouts, Validations			8
III	JAVASCRIPT: Overview, Syntax, Enabling, Placement, Variables, Operators, If-Else, Switch-Case, While Loop, For Loop, For-in Loop, Loop Control, Functions, Events, Cookies, Page Redirect, Dialog Box, Void Keyword, Page Printing, Objects, Number, Boolean, String, Arrays, Date, Math, Errors and Exceptions, Form Validation, Animation, Multimedia			7
IV	PHP: Introducing PHP, Using Variables and Operators, Controlling Program Flow, Working with Arrays, Using Functions and Classes, Working with Files and Directories, Working with Databases and SQL, Working with XML, Working with Cookies, Sessions, and Headers, Handling Errors, Securing PHP, Extending PHP.			7
Texts / References Books				
- Powell, T. A. ,HTML & CSS: The Complete Reference. - Goodman, D., Morrison, M., Novitski, P., Rayl, T. G., JavaScript Bible. - Vaswani, V., PHP: A Beginner’s Guide.				
Learning Outcomes				
- Create visually appealing and responsive web pages using HTML, CSS, and JavaScript, ensuring optimal user experience across various devices and screen sizes. - Utilize design principles and best practices to develop intuitive layouts, navigation, and user interfaces that enhance usability and accessibility. - Implement interactive features and advanced web technologies, such as front-end frameworks and APIs, to build dynamic and functional web applications.				

CSCMD101	Fundamentals of Computing	Theory	Practical	Cumulative
MD		3	0	3
		45	0	45
Course Objectives				
- Core Concepts Mastery: Introduce students to the fundamental principles and concepts of computing, including algorithms, data structures, and software design. - Programming Proficiency: Develop basic programming skills and logical thinking through hands-on experience with a programming language. - Computational Thinking: Foster computational thinking to solve problems systematically and efficiently. - Software Development Insight: Provide an understanding of the software development lifecycle and best practices in software engineering. - Ethical and Social Awareness: Instill awareness of the ethical, social, and professional issues related to computing.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Introduction to Computers: Characteristics of Computers, Evolution of Computing, Introduction to Computer Hardware/Software, Binary Number Systems, Types of Computer Software.			10
II	Problem Solving Techniques using Computers: Algorithm, Flow Charts, Pseudo code, Introduction to Low-Level, Assembly and High-Level Languages.			10
III	Introduction to Operating System: Concept, Functions, Types, Single-user/Multi-user operating system, Architectural differences Introduction to Memory: Memory Hierarchy and Management, Cache Memory, Primary Memory, Secondary Memory.			15
IV	Introduction to Computer Networks, Internet, World Wide Web: Basic of Computer networks; LAN, WAN; Concept of Internet; Applications of Internet; connecting to internet; What is ISP; Knowing the Internet; World Wide Web; Web Browsing software, Search Engines; Understanding URL; Domain name; IP Address.			10
Texts / References Books				
- Balagurusamy, E., Computing Fundamentals & C programming, TMH. - Godse, A. P., Godse, D. A. ,Fundamental of Computing and Programming (Technical Publications).				
Learning Outcomes				
- Conceptual Understanding: Students will comprehend core computing concepts such as algorithms, data structures, and programming paradigms. - Programming Skills: Students will be able to write, debug, and maintain simple programs in a programming language. - Problem-Solving Ability: Students will apply computational thinking and algorithmic approaches to solve problems. - Software Development Knowledge: Students will understand the stages of software development and the importance of testing and documentation. - Ethical Awareness: Students will demonstrate an understanding of the ethical and societal impacts of computing technologies and practices.				

Semester-II

CSCMJ201	Data Structures and Algorithms	Theory	Practical	Cumulative
		3	1	4
Major/Minor		45	30	75
Course Objectives				
- Understanding Fundamental Concepts: Grasp the key principles of data structures and their importance in computer science. - Implementation Skills: Learn to implement various data structures such as arrays, linked lists, stacks, queues, trees, and graphs. - Algorithm Efficiency: Analyze the efficiency of data structures and algorithms using Big O notation. - Problem-Solving Techniques: Develop problem-solving skills through the application of data structures in various algorithmic challenges. - Real-World Applications: Explore the practical applications of data structures in software development and system design.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Defining a Data Structure: Types of Data Structures. Linear Structures: Array, List, Stack, Queue, Applications of arrays, lists, stacks and queues.			13
II	Non-Linear Data Structures: Tree, Tree Traversals, Binary Tree, Applications of Trees, Binary Search Tree, Graph, Shortest Path, Spanning Tree, Hashing and Collision Resolution Techniques.			12
III	Introduction to Algorithm Analysis and Design: Time Complexity Analysis, Asymptotic Notations, Introduction to Design Techniques such as Greedy, Divide and Conquer, Dynamic Programming, Backtracking, Branch and Bound.			10
IV	Searching and Sorting: Linear Search, Binary Search, Bubble Sort, Selection Sort, Insertion Sort and Quick Sort.			10
Texts / References Books				
- Horowitz,E. , Sahani, S., Anderson-Freed, S., Fundamentals of Data Structures in C, Universities Press. - Tennenbaum, A. M., Langsam, Y. and Augenstein, M. J., Data Structures using C, PHI. - Knuth, D. E. ,The Art of Computer Programming (Volume I), Pearson. - Cormen, T. H. et al., Introduction to Algorithms, PHI.				
Learning Outcomes				
- Proficient in Data Structures: Ability to choose and implement the appropriate data structure for a given problem. - Enhanced Coding Skills: Improved coding proficiency through hands-on programming assignments. - Analytical Skills: Enhanced ability to analyze algorithm performance and optimize solutions. - Foundation for Advanced Topics: Establish a solid foundation for further studies in algorithms, databases, and software engineering.				

CSCSE201	Mobile Application Development	Theory	Practical	Cumulative
		2	1	3
SEC		30	30	60
Course Objectives				
• Foundational Knowledge: Understand mobile development principles, platforms, and user experience design. • Technical Proficiency: Gain hands-on experience with key programming languages and frameworks for Android. • App Integration: Learn to integrate mobile applications with backend services and APIs, focusing on data management. • Testing and Deployment: Develop skills in testing, debugging, and deploying apps to app stores, along with understanding current trends in mobile technology.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Introduction to Android: The Android Platform, Android SDK, Eclipse Installation, Android Installation, Building you First Android application, Understanding Anatomy of Android Application, Android Manifest file			8
II	Android Application Design Essentials: Anatomy of an Android applications, Android terminologies, Application Context, Activities, Services, Intents, Receiving and Broadcasting Intents, Android Manifest File and its common settings, Using Intent Filter, Permissions.			8
III	Android User Interface Design Essentials: User Interface Screen elements, Designing User Interfaces with Layouts, Drawing and Working with Animation. Testing Android applications, Publishing Android Application, Using Android preferences, Managing Application resources in a hierarchy, working with different types of resources.			7
IV	Using Common Android APIs: Using Android Data and Storage APIs, Managing data using Sqlite, Sharing Data between Applications with Content Providers, Using Android Networking APIs, Using Android Web APIs, Using Android Telephony APIs, Deploying Android Application to the World.			7
Texts / References Books				
1. Darcey, L. and Conder, S., Android Wireless Application Development, Pearson Education, 2nd ed. (2011). 2. Meier, R., Professional Android 2 Application Development, Wiley India Pvt Ltd. 3. Murphy, M. L., Beginning Android, Wiley India Pvt Ltd. 4. Burd, B., Android Application Development All in one for Dummies, Edition: I.				
Learning Outcomes				
After successful completion of this course, the students will be able to:				
• Understand about the mobile application market and web services for various mobile devices. • Understand about the various Mobile Information Design, design Tools, Mobile Platforms and Mobile Web Option. • Design the User interface with various features of Android SDK like displaying pictures, menus etc. • Utilize the messaging, networking and location based service in Android application • Create. Debug and build the apps.				

CSCMD201	Introduction to Information Technology	Theory	Practical	Cumulative
		3	0	3
MD		45	0	45
Course Objectives				
- Fundamentals Understanding: Provide students with a solid foundation in the basic concepts and principles of information technology, including hardware, software, and networking. - Technological Literacy: Develop technological literacy by familiarizing students with current IT tools and practices. - Problem-Solving Skills: Enhance problem-solving skills by teaching students how to apply IT solutions to real-world problems. - Digital Citizenship: Promote responsible digital citizenship by educating students about ethical and legal issues related to IT. - Practical Skills: Equip students with practical skills in using software applications, such as word processing, spreadsheets, and databases.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Introduction to Computers: Introduction, Definition, Characteristics of computer, Evolution of Computer, Capabilities and limitations of computer. Basic Computer Organization: Role of input-output devices in a computer system. Input Units: Keyboard, Terminals and its types. Pointing Devices, Scanners and its types, Voice Recognition Systems, Vision Input System, Touch Screen, and Output Units: Monitors and its types. Printers: Impact Printers and its types. Non Impact Printers and its types, Plotters, types of plotters, Sound cards, Speakers.			10
II	Storage Fundamentals: Primary Vs Secondary Storage, Data storage & retrieval methods. Primary Storage: RAM, ROM, PROM, EPROM, EEPROM. Secondary Storage: Magnetic Tapes, Magnetic Disks. Cartridge tape, hard disks, Floppy disks Optical Disks, Compact Disks, Zip Drive, Flash Drives. Software (S/W): Software and its needs, Types of S/W. System Software: Operating System, Utility Programs Programming Language: Machine Language, Assembly Language, High Level Language their advantages & disadvantages. Application S/W and its types: Word Processing, Spread Sheets Presentation, Graphics, DBMS S/W.			10
III	Operating System: Functions, Measuring System Performance, Assemblers, Compilers and Interpreters. Batch Processing, Multiprogramming, Multi Tasking, Multiprocessing, Time Sharing, DOS, Windows, Unix/Linux. Data Communication: Communication Process, Data Transmission speed, Communication Types (modes), Data Transmission Medias, Modem and its working, characteristics, Types of Networks, LAN Topologies, Computer Protocols, Concepts relating to networking.			15
IV	Business Data Processing: Introduction, data storage hierarchy, Method of organizing data, File Types, File Organization, File Utilities. Computer Arithmetic: Binary, Binary Arithmetic, Number System: Positional & Non Positional, Binary, Octal, Decimal, Hexadecimal, Converting from one number system to another, Converting from one number system to another , Converting from one number system to another.			10
Texts / References Books				
- Rajaraman, V., Fundamentals of Computers, PHI. - Fox, R., Information Technology, Chapman and Hall/CRC. - Sinha, P. K., & Sinha, P., Computer fundamentals. BPB publications.				
Learning Outcomes				
- Knowledge Acquisition: Students will understand the fundamental components and functions of computer systems and networks. - Skill Development: Students will be able to effectively use common software applications for personal, academic, and professional tasks. - Analytical Ability: Students will be able to analyze and solve basic IT-related problems. - Ethical Awareness: Students will demonstrate awareness of ethical, privacy, and security issues in IT. - Career Readiness: Students will be prepared for further studies or entry-level positions in IT fields, equipped with a foundational understanding of technology's role in various industries.				

Semester III

CSCMJ301	Digital Logic and Circuit Design	Theory	Practical	Cumulative
		3	1	4
Major		45	30	75
Course Objectives				
- To provide a foundational understanding of digital logic and Circuit design. - To learn number systems, Boolean algebra, logic gates, and Boolean function simplification using Karnaugh maps and the tabulation method. - To design and analyze combinational circuits, such as adders, subtractors, encoders, and multiplexers. - To design and analyze sequential circuits like flip-flops, counters, and shift registers.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Number System: Weighted and Unweighted Codes, Binary, Octal, and Hexadecimal numbers; Fixed and Floating Point Number Representations, number base conversion, Complements, Binary Arithmetic: Addition, Subtraction, Multiplication and Division, BCD Code.			10
II	Boolean algebra and Logic Gates: Introduction to Boolean algebra, laws of Boolean algebra, logic gates, universal logic gates, POS and SOP notations, Canonical logic forms. Simplification of Boolean Functions: Laws of Boolean algebra and K-Maps, Tabulation Method.			12
III	Combinational Circuits: Design Procedure of Combinational Circuits, Adders, Subtractors, Code Converters, Magnitude Comparator, Encoder, Decoder, Multiplexer, Demultiplexer, ROM.			12
IV	Sequential Circuits: Flip-Flops: SR, D, JK, T, Master/Slave F/F, Edge-triggered F/F, Excitation Tables; Registers, Counters: synchronous and asynchronous, Design of Counters, Shift Registers, RAM.			11
Texts / References Books				
- Mano, M. M. Digital logic and computer design. PHI. - Mano, M. M., & Kime, C. R..Logic and computer design fundamentals. Pearson. - Malvino, A., & Leach, D. Digital principles and applications. McGraw-Hill. - Bartee, T. C. Digital computer fundamentals. McGraw-Hill.				
Learning Outcomes				
After completion of this course the students will be able to:				
- Gain proficiency in understanding and converting between number systems such as binary, octal, and hexadecimal, and performing binary arithmetic operations. - Master the principles of Boolean algebra and logic gates, and simplify Boolean functions using Karnaugh maps and the tabulation method. - Develop skills in designing and analyzing combinational circuits, including adders, subtractors, encoders, decoders, multiplexers, etc. - Understand and design sequential circuits, including various types of flip-flops, counters, and shift registers. - Acquire competencies essential for creating efficient digital systems, crucial for developing modern electronics, computers, and embedded systems used in various practical applications in today's technology-driven world.				

CSCMJ404	Object Oriented Programming	Theory	Practical	Cumulative
Major		3	1	4
		45	30	75
Course Objectives				
• OOP Principles: Introduce the fundamental principles of object-oriented programming (OOP) and how they are implemented in Java. • Java Syntax and Semantics: Teach students the syntax and semantics of the Java programming language. • Design and Implementation: Develop skills in designing and implementing robust, reusable, and maintainable software using Java. • Problem-Solving with OOP: Enhance problem-solving abilities by applying OOP concepts to real-world scenarios. • Software Development Practices: Instill best practices in software development, including debugging, testing, and documentation.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Object Oriented Concepts: Objects and Classes, Bottom up approach, OO design principles, OO Design and Modeling.			8
II	Basic OO language Constructs: Primitive Data Types and Operations, Selection Statements, Loops, Arrays, Strings, Objects and Classes, Inheritance and method overriding, Polymorphism.			15
III	Java Language Fundamentals: Object Design: constructors, instance variables, methods. Memory models, scope, streams and I/O programming, Inner classes, Interfaces and packages, Exception Handling, Multithreading.			15
IV	Advanced Concepts: Creating GUIs and Displaying Data, Event Driven Programming. Frameworks: The framework concept, Frameworks in the Java API: Collections Framework, Graphics Framework.			7
Texts / References Books				
1. Wu, C. T. An introduction to object oriented programming with JAVA. McGraw Hill. 2. Balagurusamy, E. Programming with Java. TMH. 3. Eckel, B. Thinking in Java. Pearson. 4. Schildt, H. Java: The complete reference. McGraw Hill Education. 5. Deitel, H., & Deitel, P. Java 9 for programmers. Pearson				
Learning Outcomes				
• Understanding OOP: Students will understand and apply the core concepts of object-oriented programming, such as classes, objects, inheritance, polymorphism, and encapsulation. • Java Proficiency: Students will be proficient in writing, compiling, and executing Java programs. • Design Skills: Students will be able to design and develop object-oriented software solutions for complex problems. • Code Quality: Students will produce code that is modular, well-documented, and adheres to industry standards. • Project Experience: Students will complete projects that demonstrate their ability to apply Java and OOP concepts to build functional applications.				

CSCSE301	Python Programming	Theory	Practical	Cumulative
		2	1	3
SEC		30	30	60
Course Objectives				
- Master Python syntax, data structures, functions, and object-oriented programming to write, debug, and optimize code effectively. - Solve real-world problems by using Python for tasks like data analysis, file manipulation, and automation, leveraging libraries and frameworks.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Introduction to Computers and Programming: Introduction, Hardware and Software, How Computers Store Data, How a Program Works, Python Installation, Input, Processing, and Output: Displaying Output with the print Function, Comments, Variables, Reading Input from the Keyboard, Performing Calculations, More About Data Output.			8
II	Decision Structures and Boolean Logic: The if Statement, The if-else Statement, Comparing Strings, Nested Decision Structures and the if-elif-else Statement, Logical Operators, Boolean Variables. Repetition Structures: Introduction to Repetition Structures, The while Loop: A Condition-Controlled Loop, The for Loop: A Count-Controlled Loop, Calculating a Running Total, Sentinels, Input Validation Loops, Nested Loops			8
III	Functions: Introduction to Functions, Defining and Calling a Void Function, Designing a Program to Use Functions, Local Variables, Passing Arguments to Functions, Global Variables and Global Constants, Introduction to Value-Returning Functions: Generating Random Numbers, Writing Your Own Value-Returning Functions, Recursion, The math Module, Storing Functions in Modules Lists and Tuples: Sequences, Introduction to Lists, List Slicing, Finding Items in Lists with the in Operator, List Methods and Useful Built-in Functions, Copying Lists, Processing Lists, Two-Dimensional Lists, Tuples, Plotting List Data with the matplotlib Package.			7
IV	More About Strings: Basic String Operations, String Slicing, Testing, Searching, and Manipulating Strings. Dictionaries and Sets: Dictionaries, Sets, Serializing Objects. Files and Exceptions: Introduction to File Input and Output, Using Loops to Process Files, Processing Records, Exceptions Object-Oriented Programming: Procedural and Object-Oriented Programming, Classes, Working with Instances, Techniques for Designing Classes, Introduction to Inheritance, Polymorphism.			7
Texts / References Books				
- Gaddis, T. Starting out with Python. - Dierbach, C.Introduction to computer science using Python. - Chun, W. J. Core Python programming				
Learning Outcomes				
- Develop basic programs using fundamental structures. - Create programs using various collection data types. - Apply appropriate Python control flow structure. - Implement user defined python functions. - Understand the objected oriented concepts of Python.				

CSCMV301	IT TOOLS AND APPLICATIONS	Theory	Practical	Cumulative
Voc		3	1	4
		45	30	75
Course Objectives				
- Fundamental Understanding: Acquire a solid foundation in essential IT tools and applications used in various business environments. - Tool Proficiency: Develop practical skills in widely-used software applications, including productivity suites, project management tools, and collaboration platforms. - Problem-Solving Skills: Enhance critical thinking and problem-solving abilities by applying IT tools to real-world scenarios and case studies. - Technology Integration: Understand how to integrate different IT tools and applications to improve efficiency and streamline workflows in organizational settings.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Word Processing Word processing concepts: saving, closing, Opening an existing document, Selecting text, Editing text, Finding and replacing text, printing documents, Creating and Printing Merged Documents, Character and Paragraph Formatting, Page Design and Layout. Editing and Profiling Tools: Checking and correcting spellings, Handling Graphics, Creating Tables and Charts, Document Templates and Wizards. Applications in different domains.			12
II	Spreadsheet Package Spreadsheet Concepts, Creating, Saving and Editing a Workbook, Inserting, Deleting Work Sheets, entering data in a cell / formula Copying and Moving from selected cells, handling operators in Formulae, Functions: Mathematical, Logical, statistical, text, financial, Date and Time functions, Using Function Wizard. Formatting a Worksheet: Formatting Cells – changing data alignment, changing date, number, character or currency format, changing font, adding borders and colors, Printing worksheets, Charts and Graphs – Creating, Previewing, Modifying Charts. Integrating word processor, spread sheets, web pages. Applications in different domains.			14
III	Presentation Package Creating, Opening and Saving Presentations, Creating the Look of Your Presentation, Working in Different Views, Working with Slides, Adding and Formatting Text, Formatting Paragraphs, Checking Spelling and Correcting Typing Mistakes, Making Notes Pages and Handouts, Drawing and Working with Objects, Adding Clip Art and other pictures, Designing Slide Shows, Running and Controlling a Slide Show, Printing Presentations. Applications in different domains.			10
IV	Data Base Operations Data Manipulation-Concept: Database, Relational Database, Integrity. Operations: Creating, dropping, manipulating table structure. Manipulation of Data: Query, Data Entry Form, Reports. Applications in different domains.			9
Texts / References Books				
- Sinha P.K., and Sinha P., “Foundations of Computing”, BPB Publication. - Sagman S, “MS Office for Windows XP”, Pearson Education. - ITL Educational Society, “Introduction to IT”, Pearson Education. - Miller M, “Absolute Beginners Guide to Computer Basics”, Pearson Education.				
Learning Outcomes				
At the end of the course the students will be able to: - Acquire the foundation level knowledge required to understand computer and its operations. - Understand to use the packages of word processing, spread sheet and presentation in detail. - Understand the issues related to IT and IT applications.				

CSCMD301	Introduction to Cyber Security	Theory	Practical	Cumulative
		3	0	3
MD		45	0	45
Course Objectives				
- To prepare students with the technical knowledge and skills needed to protect and defend computer systems and networks. - To develop graduates that can plan, implement, and monitor cyber security mechanisms to help ensure the protection of information technology assets. - To develop graduates that can identify, analyze, and remediate computer security breaches.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Introduction History of Internet, Cyber Crime, Information Security, Computer Ethics and Security Policies, Guidelines to choose web browsers, Securing web browser, Antiviruses, Email security, Guidelines for setting up a Secure password, Two-steps authentication ,Password Manager, Wi-Fi Security.			15
II	Social media, Smart Phone and basic Windows security: Guidelines for social media security, best practices for safer Social, Introduction to mobile phones, Smartphone Security, Android Security, IOS Security, Basic Security for Windows User Account Password, etc			10
III	Cyber Security Initiatives in India and online Banking Security: IT Security Act, Cyber Security Initiatives in India, Cyber Security Exercise, Cyber Security Incident Handling.			10
IV	Cyber Security Assurance, Online Banking Security, Mobile Banking Security, Security of Debit and Credit Card , UPI Security, Security of Micro ATMs, e-wallet Security Guidelines, Security Guidelines for Point of Sales(POS).			10
Texts / References Books				
- Cryptography and Network Security Principles and Practice, William Stallings, Pearson. - Computer Security Fundamentals, by William (Chuck) Easttom II, Pearson.				
Learning Outcomes				
- Understand what is cyber security - Understand various aspects of Internet Security. - Know the cyber security initiatives taken by the Government - Know how to do online transactions in a secured way				

Semester –IV

CSCMJ401	Mathematics For Computer Science	Theory	Practical	Cumulative
		3	0	3
Major		45	0	45
Course Objectives				
- To use sets for solving applied problems, and use the properties of set operations algebraically. Work with relations and investigate their properties. Investigate functions as relations and their properties. - To learn about partial ordering , lattice theory, and theoretical foundation of Boolean algebra. - To introduce concepts of mathematical logic – propositional and predicate - for analyzing propositions and proving theorems. - To learn rule of inference in propositional and predicate framework of logic. - To learn basics concept of probability. - To introduce basic concepts of graphs, types of graphs and its application to solve real life problem.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Set, set operations, relation, matrix and graph representation of relation, Properties of relation, and Partial order relations. Hasse diagram, lattice, and Boolean algebra. Number theoretic function, and recursion.			10
II	Propositional calculus / statement calculus – statement, logic connective, statement formula and truth table, well formed formula, tautology, contradiction, substitution instance, equivalence of formula, duality law, tautological implication, functionally complete set of connectives, satisfiability, decision problem, normal forms. Theory of inference for propositional calculus. Predicate calculus: Limitation of propositional calculus, statement representation using predicate and quantifier.			15
III	Probability: Sample Space and Events, Axioms of Probability, Conditional Probability. Random Variables and Expectations: Random variables, Jointly Distributed Random variables, Expectation, Variance, Co-variance, Probability Distributions. Parameter Estimation-Maximum Likelihood Estimates; Regression Analysis; Applications, Markov Process, Poisson Process.			10
IV	Basic Concepts of Graphs and Trees, Adjacency and Incidence Matrices, Planar Graphs, Graph Coloring, Eulerian and Hamiltonian graphs, Applications of Graph Theoretic Concepts to Computer Science.			10
Texts / References Books				
J. P. Trembley and R. P. Manohar, Discrete Mathematical Structures with Applications to Computer Science, McGraw Hill. K. Rosen, Discrete Mathematics, Tata McGraw Hill. - Irwin Miller, Maryless Miller, Mathematical Statics with Applications, PHI. - Oliver C. Ibe, Fundamentals of Applied Probability and Random Processes, Academic Press, 2014 - Jay Devore, Probability and Statistics for Engineering and the Sciences, Cengage Learning India, 2022.				
Learning Outcomes				
After successful completion of this course, students will be able to:				
- Understand sets and perform operations and algebra on sets. - Determine properties of relations identify equivalence and partial order relations, sketch relations, functions and determine their properties; understand the theoretical foundation of Boolean algebra. - Analyze logical propositions via truth tables, able express proposition in different standard form. Use inference rules to solve different reasoning problems. - Learn basic concepts of probability and statistics, and able to use parameter estimation and regression for real life problem. - Define graphs, digraphs and trees, and identify their main properties. Learn to model the problem using and their solution.				

CSCMJ402	Database Management System	Theory	Practical	Cumulative
		3	1	4
Major		45	30	75
Course Objectives				
- Introduce fundamental concepts, terminology and application of databases. - To study and understand the design concepts and creation of relational databases. - To study and understand the basic and advanced SQL commands. - Provide overview of transaction management, concurrency control, database recovery and security. - Provide an overview of advance database technologies.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Introduction: Database Systems, View of Data Models, Database Languages, DBMS Architecture, Database Users and Data Independence. ER Modeling: relation types, role and Structural Constraints, Extended ER Modeling Features, Design of an ER Database Schema, Reduction of ER Schema to Tables.			15
II	Relational Model: Relational Model Concepts, Relational Algebra. Introduction to SQL: SQL data types and literals, Types of SQL commands, SQL operators, Tables, views and indexes, Queries and sub queries, Aggregate functions.			10
III	Relational Database Design: Functional and Multi-valued Dependencies, Desirable Properties of Decomposition, Normalization up to 3 NF and BCNF.			10
IV	Transaction Management, Concurrency Control, Recovery Techniques, Security, Introduction to Query Processing and Query Optimization. Data warehousing, Data mining, Distributed databases, Big Data and NoSQL.			10
Texts / References Books				
- Date, C. J., An Introduction to Database Systems, AddisonWesley. - Silberschatz, A., Korth, H. F. and Sudharshan, S., Data Base System Concepts, McGrawHill. - Ullman, J. D., Principles of Database Systems, Galgotia. - Elmasri, R., Navathe, S.B., Fundamentals of Database Systems, Pearson Education Asia. - Ramakrishnan, R., Database Management Systems, McGraw-Hill Education				
Learning Outcomes				
After successful completion of this course, students will be able to: - Describe the fundamental elements of relational database management systems - Explain the basic concepts of relational data model, entity-relationship model, relational database design, relational algebra and SQL. - Design ER-models to represent simple database application scenarios - Convert the ER-model to relational tables, populate relational database and formulate SQL queries on data. - Improve the database design by normalization. - Understand the concept of a database transaction and related database facilities, including concurrency control, backup and recovery, and data object locking and protocols. - Compare, contrast and analyze the various emerging technologies for database systems such as NoSQL.				

CSCMJ403	Computer Organization	Theory	Practical	Cumulative
		3	0	3
Major		45	0	45
Course Objectives				
• To study and understand the basic structure and operation of digital computer. • To study and understand the design of arithmetic and logic unit and implementation of fixed point and floating-point arithmetic operations • To study and understand the two types of control unit techniques. • To study and understand the hierarchical memory system including cache memories and virtual memory • To study and understand the different ways of communicating with I/O devices and standard I/O interfaces				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Introduction: Components of a Computer System, Machine Instruction: Opcode and Operand, Machine’s Instruction Cycle, Organization of Central Processing Unit: ALU, Control Unit and Registers (Special and General Purpose), Memory, and Input/Output.			12
II	Memory Organization: Memory Hierarchy, Cache Memory: Cache Performance, Cache Mapping Techniques (Direct, Associative, and Set-Associative) and Locality of Reference, Main Memory (DRAM and ROM), Secondary Memory, Virtual Memory, and Characteristics of Different Types of Memory.			12
III	CPU Organization: Instruction Formats, Instruction Types, Addressing Mode and its Types, Common Microprocessor Instructions, RISC and CISC, Physical Cores, Logical Cores, and Multithreading.			12
IV	I/O Organization: Peripheral devices, I/O interface, Modes of Transfer, Priority Interrupt, Direct Memory Access, Modes of Data Transfer in DMA, Input-Output Processor, and Serial Communication.			9
Texts / References Books				
1. Mano, M. M. Computer system architecture. PHI. 2. Rajaraman, V., & Radhakrishnan, T. An introduction to digital computer design. PHI. 3. Stallings, W. Computer organization and architecture: Designing for performance. Prentice Hall.				
Learning Outcomes				
The students are expected to be able to demonstrate the following knowledge, skills and attitudes after completing this course • To Identify and Describe Computer Components • A detailed understanding of architecture and functionality of central processing unit. • A depth understanding of Memory Hierarchy and various types of memory communicating with processing unit. • Understanding of I/O devices communicating with Processing Unit and also knowing the characteristics of multi processors. • To measure and analyze the performance of computer systems using appropriate metrics.				

CSCMJ404	Numerical Computing	Theory	Practical	Cumulative
		3	1	4
Major/Minor		45	30	75
Course Objectives				
- To provide a comprehensive understanding of numerical methods essential for computational problem-solving. - To understand errors in computer arithmetic and normalization techniques. - To solve equations using bisection, false position, and Newton-Raphson methods. - To explore iterative methods for systems of linear equations, matrix inversion, eigenvalues, and eigenvectors, with a focus on diagonalization using Jacobi’s method. - To include polynomial interpolation, numerical differentiation, and integration techniques such as Trapezoidal and Simpson’s Rules. - To address numerical solutions to differential equations using methods like Euler’s, Runge-Kutta, etc.. - To Equip students with skills critical for developing efficient algorithms and addressing complex computational challenges in today’s technology-driven world.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Errors in Computer Arithmetic, Normalization. Bisection, False position and Newton-Raphson methods for solution of nonlinear equations. Errors in the solutions, Convergence of Solutions.			10
II	Gauss, Gauss-Siedel and Iterative methods for system of linear equations. Ill conditioned system, Pivotal Condensation, Matrix Inversion, Eigen-values, Eigen-vector, Diagonalization of Real Symmetric Matrix by Jacobi’s Method.			10
III	Introduction to Finite Differences, Polynomial Interpolation using Newton’s and Lagrange’s formulae.Numerical Differentiation: Numerical Integration: Trapezoidal Rule, Simpson’s Rule, Weddle’s Rule, Gauss Quadrature Formula. Error in numerical Integration.			15
IV	Numerical Solution of differential Equations: Picards Method, Taylor’s Series Method, Euler’s Method, Modified Euler’s Method, Runge-Kutta Method, Predictor-Corrector Method.			10
Texts / References Books				
- Rajaraman, V. Computer oriented numerical methods. PHI. - Acton, F. Numerical methods that work. Harper and Row. - Conte, S. D., & Boor, C. D. Elementary numerical analysis. McGraw-Hill. - Sastry, S. S. Introductory methods of numerical analysis. PHI. - Gerald, C. F., & Wheatley, P. O. Applied numerical analysis. Addison Wesley.				
Learning Outcomes				
After successful completion of this course, student will be able to - Gain proficiency in identifying and mitigating errors in computer arithmetic and normalization techniques. - Master methods for solving nonlinear equations, including bisection, false position, and Newton-Raphson, with an understanding of convergence and error analysis. - Develop skills in iterative methods for systems of linear equations, matrix inversion, and eigen-decomposition, including Jacobi’s method for diagonalization. - Effectively use polynomial interpolation techniques and numerical integration methods such as Trapezoidal and Simpson’s Rules. - Become adept at solving differential equations using Picard’s Method, Euler’s Method, Runge-Kutta Method etc.. - Acquire skills crucial for addressing real-world computational problems and developing efficient algorithms.				

Semester –V

CSCMJ501	Operating System Concepts	Theory	Practical	Cumulative
		3	1	4
Major		45	30	75
Course Objectives				
- To learn the mechanisms of OS to handle processes and threads and their Communication - To learn the mechanisms involved in memory management in contemporary OS - To gain knowledge on distributed operating system concepts that includes architecture, Mutual exclusion algorithms, deadlock detection algorithms and agreement protocols - To know the components and management aspects of concurrency management				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Introduction: Definition, Design Goals, Evolution; Batch processing, Multiprogramming, Timesharing; Real-time operations; Structure and Functions of Operating System. Process Management: Process states, State Transitions, Process Control Structure, Context Switching, Process Scheduling, Threads.			10
II	Memory Management: Address Binding, Dynamic Loading and Linking Concepts, Logical and Physical Addresses, Contiguous Allocation, Fragmentation, Paging, Segmentation, Combined Systems, Virtual Memory, Demand Paging, Page fault, Page replacement algorithms, Global Vs Local Allocation, Thrashing, Working Set Model.			15
III	Concurrent Processes: Process Interaction, Shared Data and Critical Section, Mutual Exclusion, Busy form of waiting, Lock and unlock primitives, Synchronization, Classical Problems of Synchronization, Semaphores, Monitors, Conditional Critical Regions, System Deadlock, Wait for Graph, Deadlock Handling Techniques: Prevention, Avoidance, Detection and Recovery.			10
IV	File and Secondary Storage Management: File Attributes, File Types, File Access Methods, Directory Structure, Allocation Methods, Free Space management; Disk Structure, Logical and Physical View, Disk Head Scheduling.			10
Texts / References Books				
A. Silberschatz, P. B. Galvin, G. Gagne, Operating System Concepts, Addison Wesley. W. Stalling, Operating Systems: Internals and Design Principles, PHI. A. S. Tanenbaum, Modern operating Systems, PHI.				
Learning Outcomes				
After the completion of this course the students will be able to:				
- Interpret Operating System Structure, Operations, Services and Process - Elaborate Multithreaded Programming, Process Scheduling and Synchronization - Evaluate different memory management schemes - Design and implement File system functionalities - Experiment with various disk management schemes				

CSCMJ502	Design and Analysis of Algorithms	Theory	Practical	Cumulative
Major		3	1	4
		45	30	75
Course Objectives				
• The goal of this course is to provide a solid background in the design and analysis of the major classes of algorithms. • At the end of the course students will be able to develop their own versions for a given computational task and to compare and contrast their performance. • Students are also expected to develop advanced level of algorithms with low complexity.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Introduction to Algorithms: Time and Space complexity, Average and Worst-case analysis, Asymptotic notation, Master theorem Recurrence equations and their solution.			10
II	Algorithmic Techniques: Search techniques (backtracking and bounding), Search Trees, Sorting algorithms – Heapsort, Quick sort, Sorting in linear time (radix sort, bucket sort).			12
III	Algorithmic Paradigms: Greedy algorithms (Activity-selection problem, Huffman coding, Knapsack, Shortest path and Minimum Spanning tree in graphs), Divide and Conquer – Merge Sort, Integer Multiplication, Solving Recurrence-substitution method and recursion-tree,; Dynamic programming (0/1 Knapsack Problems, Traveling Salesman problem, Matrix multiplication, Shortest paths algorithms,, Optimal binary search trees).			12
IV	Computational complexity: Problem classes: P, NP, NP-complete, NP-hard, Examples of NP-complete problems, Cook's theorem.			11
Texts / References Books				
1. T. H. Cormen, C. E. Leiserson, R. L. Rivest , C. Stein, Introduction to Algorithms, PHI. 2. M. A. Weiss, Data Structures and Problem Solving Using Java, Addison Wesley. 3. A. Aho, V. Alfred, J. Hopcroft, J. D. Ullman, The Design and Analysis of Computer Algorithms, Addison Wesley. 4. J. Kleinberg and E. Tardos, Algorithm Design, Pearson, 2006. 5. E. Horowitz, S. Sahni, S. Rajasekaran, Fundamentals of Computer Algorithms, Galgotia.				
Learning Outcomes				
After the completion of this course the students will be able to:				
• Given an algorithm, identify the problem and solves to problems with efficient algorithms. • Write algorithms to choosing the best one or a combination of two or more of the algorithm design techniques: Iterative, divide-n-conquer, Greedy, Dynamic Programming using appropriate data structures. • To explore real life applications with efficient solutions.				

CSCMJ503	Theory of Computation	Theory	Practical	Cumulative
		3	1	4
Major		45	30	75
Course Objectives				
- To provide a comprehensive understanding of the Theory of Computation, focusing on fundamental concepts and their applications. - To explore finite state automata, including deterministic and nondeterministic types, regular expressions, and their equivalence with finite state machines. - To study context-free grammars, pushdown automata, their properties, and techniques for simplifying grammars and understanding derivation trees. - To cover Turing machines, their variants, the Chomsky hierarchy, and concepts of computability. - To examine decidability issues, including Post's correspondence problem, to understand the limits of computation and address key challenges in modern computing systems.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Alphabets, string, language, basic operations on language, concatenation, union, Kleene star. Finite Automata (FA): Non deterministic FA and deterministic FA, NFA with ϵ - moves, Regular Expressions, Equivalence of regular expression and FA, Pumping lemma, closure properties, FA minimization: Myhill-Nerode theorem and Equivalence theorem, Formal Language and Grammar: Production systems, Finite automata with output, Regular grammar.			15
II	Context free grammars, Normal forms, Simplification of CFG, Derivation trees and ambiguity. Pushdown automata: Acceptance by empty store and final state, Deterministic pushdown automata, Equivalence between pushdown automata and context-free grammars, Pumping lemma for CFL, Closure properties of CFL.			10
III	Turing machine as a model of computation, configuration of simple Turing machine, Church Turing Thesis, Universal Turing Machine, decidability, halting problem, context sensitive languages and linear bounded automata, Chomsky Hierarchy.			10
IV	Undecidability: A Language that is not Recursively Enumerable (RE), an Undecidable Problem that is RE, Undecidable Problems about Turing Machine. Post's Correspondence Problem, The Classes P and NP.			10
Texts / References Books				
- Hopcraft, J. E., Motwani, R., & Ullman, J. D. (n.d.). Introduction to automata theory, languages and computation. Pearson. - Linz, P. An introduction to formal language and automata. Narosa Publisher. - Mishra, K. L. P., & Chandrasekaran, N. Theory of computer science: Automata, languages and computation. PHI. - Martin, T. C. Theory of computation. Tata McGraw-Hill.				
Learning Outcomes				
After successful completion of this course, students will be able to:				
- Gain a deep understanding of finite automata, including deterministic and nondeterministic types, and their equivalence with regular expressions, as well as concepts like the Pumping Lemma and closure properties. - Master context-free grammars, pushdown automata, and their relationships, including normal forms, derivation trees, and ambiguity. - Achieve proficiency in Turing machines, covering their construction, variations, and the Chomsky hierarchy, as well as concepts of computability. - Analyze decidability issues, such as Post's correspondence problem, equipping students with critical skills to address theoretical and practical challenges in computation.				

CSCMV501	Introduction to Multimedia	Theory	Practical	Cumulative
		3	1	4
Minor(VOC-2)		45	30	75
Course Objectives				
- To understand Multimedia Concepts - To apply Multimedia Tools - To implement Multimedia Design Principle - To explore Emerging Trends				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Introduction to Multimedia What is multimedia, Components of multimedia, Web and Internet multimedia applications, Transition from conventional media to digital media. Computer Fonts and Hypertext: Usage of text in Multimedia, Families and faces of fonts, outline fonts, bitmap fonts International character sets and hypertext, Digital fonts techniques.			12
II	Audio fundamentals and representations Digitization of sound, frequency and bandwidth, decibel system, data rate, audio file format, Sound synthesis, MIDI, wavetable, Compression and transmission of audio on Internet, Adding sound to your multimedia project, Audio software and hardware.			12
III	Image fundamentals and representations Colour Science , Colour Models, Colour palettes, Dithering, 2D Graphics, Image Compression and File Formats :GIF, JPEG, JPEG 2000,PNG, TIFF, EXIF, PS, PDF, Basic Image Processing [Can Use Photoshop], Use of image editing software, White balance correction, Dynamic range correction, Gamma correction, Photo Retouching.			11
IV	Video and Animation Video Basics, How Video Works, Broadcast Video Standards, Analog video, Digital video, Video Recording and Tape formats, Shooting and Editing Video (Use Adobe Premier for editing), Video Compression and File Formats. Video compression based on motion compensation, MPEG-1, MPEG-2, MPEG-4, MPEG-7, MPEG-21, Animation: Cell Animation, Computer Animation, Morphing.			10
Texts / References Books				
1. Tay Vaughan, “Multimedia making it work”, Tata McGraw-Hill, 2008. 2. Rajneesh Aggarwal & B. B Tiwari, “ Multimedia Systems”, Excel Publication, New Delhi, 2007. 3. Li & Drew, “Fundamentals of Multimedia”, Pearson Education, 2009.				
Learning Outcomes				
- Demonstrate proficiency in using multimedia software and tools for creating and editing text, images, audio, and video. - Design and create multimedia content that integrates multiple media types effectively. - Identify and utilize various multimedia file formats and understand their appropriate use cases.				

Semester VI

CSCMJ601	Software Engineering	Theory	Practical	Cumulative
		4	0	4
Major		60	0	60
Course Objectives				
- To study and understand software engineering methods and practices, and their appropriate application. - To study and understand Software life-cycle models, software requirements and the SRS documents. - To study and understand the role of project management including planning, scheduling, risk management, etc. - To study and understand software architecture and design - To study and understand software testing approaches. - To study and understand software measurement and software risks. - To study and understand on quality control and how to ensure good quality software.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Introduction: Software crisis, Software development Projects, Exploratory software development, Emergence of Software Engineering, Software life-cycle models: Waterfall model and its extensions, Rapid Application Development (RAD), Agile Development, Spiral Model etc. CMM, Software Quality, role of metrics and measurement. Requirements Analysis and Specification: SRS Building Process, Specification Languages, Validation of SRS, metrics, monitoring and control, Object Oriented analysis.			15
II	Software Project Management: Responsibilities & Complexities of Software project Management, Project Planning, Matrices for Project size estimation, Project Estimation techniques: Empirical, Heuristic, and analytical Estimation techniques, Project Scheduling techniques: Activity Network, CPM, PERT Chart, Gantt Chart, Organization & Team Structure, Risk Management, and Configuration Management.			15
III	Software Architecture & Design: Role of Software Architecture, Architecture Views, Component and Connector View, Architecture Styles for C&C View, Architecture Evaluation. Software Design and Implementation: Overview of the design Concepts, Cohesion and Coupling, Software design Approaches: Function Oriented and Object-Oriented design concepts, Design Metrics, Design Translation Process.			15
IV	Software Testing and Reliability: Basic concepts & terminologies of coding and testing, Techniques of testing, Debugging approaches, Software Maintenance, Software Reliability and Availability Models & Metrics, Software Reengineering, Software Reuse. Introduction to IEEE Standards, Case Studies.			15
Texts / References Books				
P. Jalote, An Integrated Approach to Software Engineering, Illrd Edition, Narosa Publishing House. R. S. Pressman, Software Engineering: A Practitioner’s approach, McGraw-Hill. I. Sommerville, Software Engineering: Pearson Education. C. Ghezzi, M. Jazayeri, D. Mandrioli, Fundamentals of Software Engineering, PHI. R. Mall, Fundamentals of Software Engineering, PHI. - RN Taylor, N Medvidovic, EM Dashofy (2009) Software Architecture: Foundations, Theory and Practice, John Wiley.				
Learning Outcomes				
- How to apply the software engineering lifecycle by demonstrating competence in communication, planning, analysis, design, construction, and deployment - An ability to work in one or more significant application domains - Work as an individual and as part of a multidisciplinary team to develop and deliver quality software - Demonstrate an understanding of and apply current theories, models, and techniques that provide a basis for the software lifecycle - Demonstrate an ability to use the techniques and tools necessary for engineering practice				

CSCMJ501	Compiler Design	Theory	Practical	Cumulative
		3	1	4
Major		45	30	75
Course Objectives				
- To provide an understanding of compiler design and construction. - To cover the phases of a compiler and the tool-based approach to compiler development. - To explore lexical analysis, including tokenization, lexemes, patterns, and error reporting using tools like LEX. - To study syntax analysis, including both top-down and bottom-up parsing with tools such as YACC. - To address type checking, type systems, and runtime system components, including storage organization and symbol tables. - To focus on intermediate code generation, code optimization techniques, and the generation of efficient machine code through various optimization strategies.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Introduction: Translators, Various phases of compiler, tool based approach to compiler construction. Lexical analysis: token, lexeme and patterns, difficulties in lexical analysis, error reporting, implementation, regular definition, transition diagrams, LEX.			10
II	Syntax Analysis:Top Down parsing (recursive descent parsing, predictive parsing), operator precedence parsing, bottom-up parsing (SLR, LALR, Canonical LR), YACC. Syntax directed definitions: inherited and synthesized attributes, dependency graph, evaluation order, bottom-up and top-down evaluation of attributes, L-attributed and S attributed Definitions.			15
III	Type checking: type system, type expressions, structural and name equivalence of types, type conversion, overloaded functions and operators, polymorphic functions. Run time system: storage organization, activation tree, activation record, parameter passing, dynamic storage allocation, symbol table: hashing, linked list, tree structures.			10
IV	Intermediate code generation: Intermediate Representation, Translation of Declarations, Assignments, Control Flow, Boolean Expressions and Procedure Calls. Code generation and Optimization: Basic blocks and flow graphs, code generation, DAG representation of programs, code generation from DAGs, Loop Optimization, Global Optimization, Loop Optimization, Peep-Hole optimization.			10
Texts / References Books				
- Aho, A. V., & Ullman, J. D. (1977). Principles of Compiler Design (Addison-Wesley series in computer science and information processing). Addison-Wesley Longman Publishing Co., Inc. - Tremblay, J. P., & Sorenson, P. G. (1985). Theory and Practice of Compiler Writing. McGraw-Hill, Inc. A. Holub, (2023) Compiler Design in C, PHI.				
Learning Outcomes				
After successful completion of this course, students will be able to:				
- Understand the phases of compiler construction and the principles behind lexical analysis, including tokenization, pattern matching, and error reporting. - Develop proficiency in syntax analysis through both top-down and bottom-up parsing methods. - Grasp concepts related to type checking, type systems, and runtime systems, including storage organization and symbol tables. - Acquire skills in intermediate code generation and apply optimization techniques, such as loop and global optimizations, to produce efficient machine code.				

CSCMJ602	Computer Networks	Theory	Practical	Cumulative
Major		3	1	4
		45	30	75
Course Objectives				
- To provide students with an overview of the concepts and fundamentals of data communication and computer networks. - To introduce students to local, metropolitan and wide area networks use the standard OSI reference model as a framework and to the Internet protocol suite and network tools and programming using various networking technologies.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Introduction:Network Definition; Network Topologies; Network Classifications; Layered Network Architecture; Overview of OSI Reference Model; Overview of TCP/IP Protocol Suite. Physical Layer:Analog and Digital Signals; Digital to Digital Conversion Schemes; Analog to Digital Conversion Schemes; Parallel and Serial transmission; Digital to Analog Conversion Schemes; Analog to Analog Conversion Schemes; Multiplexing and Spread Spectrum; Transmission Media; Circuit Switching and Packet Switching.			15
II	Data Link Layer: Error Detection and Error Correction Techniques; Data Link Control- Framing and Flow control; Error Recovery Protocols- Stop and Wait ARQ, Go-Back-N ARQ, Selective-Repeat ARQ; HDLC; Point to Point Protocol. Multiple Access: Random Access Protocols; Controlled Access Protocols; Channelization - FDMA, TDMA and CDMA.			10
III	Networks Layer: Logical Addressing; Internet protocols; Address Mapping and Error Reporting; Delivery, Forwarding, and Routing.			10
IV	Transport Layer:Process-to-Process Delivery: UDP, TCP, and SCTP; Congestion Control and Quality of Service. Application Layer:Domain Name System; Remote Logging, Electronic Mail, and File Transfer; WWW, HTTP, and SNMP.			10
Texts / References Books				
1. B. A. Forouzan: Data Communications and Networking, Fourth edition, THM. 2. A. S. Tanenbaum: Computer Networks, Fourth edition, PHI. 3. Douglas E. Comer: Computer Networks and Internets, Pearson.				
Learning Outcomes				
- Inspect the basics of data communication and various categories of networks - Identify the technologies for error free transmission of data over internet - Apply various routing protocols to select optimal path and relate addressing entities in Network Layer - Implement different transport and application layer protocols which enables data communication over internet - Configure intermediate devices used in networks				

CSCMJ604	Artificial Intelligence	Theory	Practical	Cumulative
		3	1	4
Major		45	30	75
Course Objectives				
- To introduce the basic principles in artificial intelligence research by covering simple representation schemes, problem solving paradigms, constraint propagation and search strategies. - To explore the students with the areas of AI application such as knowledge representation, natural Language processing and expert systems.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	INTRODUCTION: Intelligent Agents – Agents and environments - Good behavior - The nature of environments- Structure of agents- Problem Solving - Problem solving agents- Example problems- Searching for solutions- Uniformed search strategies- Avoiding repeated states			11
II	SEARCHING TECHNIQUES: Informed search strategies- Heuristic function- Local search algorithms and optimistic problems- Local search in continuous spaces- Online search agents and unknown environments- Constraint satisfaction problems (CSP)- Backtracking search and Local search for CSP – Structure of problems- Adversarial Search – Games- Optimal decisions in games – Alpha – Beta Pruning			11
III	KNOWLEDGE REPRESENTATION: First order logic – Representation revisited- Syntax and semantics for first order logic Using first order logic-Knowledge engineering in first order logic- Inference in First order logic – prepositional versus first order logic- Unification and lifting- Forward chaining – backward chaining- Resolution			12
IV	KNOWLEDGE IN LEARNING: Learning from observations - Forms of learning - Inductive learning-Learning decision trees - Ensemble learning- Knowledge in learning – Logical formulation of learning – Explanation based learning – Learning using relevant information – Inductive logic programming- Statistical learning methods- Learning with complete data- Learning with hidden variable - EM algorithm			11
Texts / References Books				
1. Artificial Intelligence – A Modern Approach, Stuart Russell, Peter Norvig, 3rd Edition, Pearson Education, 2015. 2. Artificial Intelligence, Elaine Rich and Kevin Knight, 2nd Edition, Tata McGraw-Hill, 2003. 3. Artificial Intelligence-Structures and Strategies for Complex Problem Solving, George F. Luger, Pearson Education 2002.				
Learning Outcomes				
- Explain the foundations and history of Artificial Intelligence, as well as the science of agent design. - Illustrate the use of problem-solving techniques, such as the various search methods, games, and constraint satisfaction problems. - Demonstrate AI’s use of knowledge representation, through logic agents and first-order logic to address AI problems. - Design simple software to experiment with various AI learning concepts and analyze results. - Build self-learning and research skills to be able to tackle a topic of interest on his/her own or as part of a team.				

CSCMV601	Data Analysis and Visualization	Theory	Practical	Cumulative
VOC		3	1	4
		45	30	75
Course Objectives				
- The objective of this course is to equip students with the skills and knowledge necessary to perform comprehensive data analysis and visualization using Python. - Students will learn to navigate the data science process, utilize essential Python libraries, handle and clean diverse data sets, and apply advanced techniques for data wrangling and visualization.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Introduction to Data Science and Python: Overview of Data Science, Exploratory Data Analysis (EDA), and the Data Science Process. Introduction to Python for Data Analysis: Python Shell (iPython), Jupyter Notebook, and key Python libraries (NumPy, pandas, matplotlib, SciPy, scikit-learn, statsmodels).			10
II	Data Handling with Pandas: Basics of Pandas: Data structures, arrays, vectorized computation, and essential functionality. Data Operations: Summarizing data, computing descriptive statistics, and handling data loading, storage, and file formats. Data Handling Techniques: Reading/writing text data, web scraping, binary formats, interacting with web APIs, and databases. Data Cleaning: Managing missing data, data transformation, and string manipulation.			15
III	Data Wrangling and Visualization: Data Wrangling: Hierarchical indexing, combining and merging datasets, reshaping, and pivoting. Data Visualization: Basics of matplotlib, plotting with pandas and seaborn, and other visualization tools.			10
IV	Advanced Data Analysis: Data Aggregation: Group by mechanics, aggregation techniques, split-apply-combine, and pivot tables. Time Series Analysis: Handling date/time data, time series basics, date ranges, frequencies, shifting, time zones, and resampling. Advanced Pandas: Working with categorical data, advanced GroupBy operations, and method chaining techniques.			10
Texts / References Books				
1. McKinney, W.(2017). Python for Data Analysis: Data Wrangling with Pandas, NumPy and IPython. 2nd edition. O’Reilly Media. 2. O’Neil, C., & Schutt, R. (2013). Doing Data Science: Straight Talk from the Frontline O’Reilly Media.				
Learning Outcomes				
After successful completion of this course, students will be able to:				
- Students will be skilled in using Python and Jupyter Notebook with essential libraries to analyze and visualize data effectively. - They will know how to clean and transform data from various sources, manage missing values, and prepare data for analysis. - Students will be able to combine, reshape, and organize data as needed, and will be proficient in creating and interpreting visualizations using tools like matplotlib and seaborn. - They will have the capability to perform advanced data tasks, including data aggregation and time series analysis.				

Semester- VII

CSCMJ701	Machine Learning	Theory	Practical	Cumulative
Major/Minor		3	1	4
		45	30	75
Course Objectives				
- Foundational Knowledge: Introduce students to the fundamental concepts and principles of machine learning, including supervised, unsupervised, and reinforcement learning. - Algorithm Understanding: Provide a deep understanding of various machine learning algorithms and their appropriate applications. - Practical Skills: Develop practical skills in implementing and evaluating machine learning models using programming languages and libraries. - Data Handling: Teach students techniques for data preprocessing, feature engineering, and data visualization. - Ethical Considerations: Discuss ethical considerations, fairness, and biases in machine learning models and their societal impacts.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Supervised Learning (Regression/Classification): Overview for ML, Supervised Learning, Unsupervised Learning, Problems, Data, and Tools. Linear Regression, Polynomial Regression, Features, Scaling, Cost Function, Gradient Descent, Learning Rate. Distance- based methods, Nearest-Neighbours, Decision Trees, Naïve Bayes, Linear Regression, Logistic Regression, Generalized Linear Models-Support Vector Machines, Nonlinearity and Kernel Multi-class/Structured Outputs, Ranking			10
II	Unsupervised Learning: K-means Clustering, PCA and kernel PCA -Matrix Factorization and Matrix Completion-Generative Models (mixture models and latent factor models), Expectation Maximization, Dimensionality Reduction.			10
III	Performance Evaluation: Evaluating Machine Learning algorithms and Model Selection, Performance Measure, Error Analysis, Confusion Matrix, Precision and Recall Tradeoff, F1 Score, Macro F1, Accuracy, Skewed Classes Introduction to Statistical Learning Theory, Ensemble Methods (Boosting, Bagging, Random Forests).			15
IV	Modelling and Applications: Sparse Modeling and Estimation, Modeling Sequence/Time-Series Data, Feature Representation Learning. Discriminative vs. Generative Models, Marginalization, Conditioning, Normalization, and Conditional Independence, Bayes Theorem, Markov Random Field, Naïve Bayes Model, Decision Tree, Recent trends in various learning techniques of machine learning and classification methods for IOT applications.			10
Texts / References Books				
1. Tom Mitchell, “Machine Learning”, McGraw-Hill, 1997. 2. Kevin Murphy, Machine Learning: A Probabilistic Perspective, MIT Press, 2012 3. Trevor Hastie, Robert Tibshirani, Jerome Friedman, The Elements of Statistical Learning, Springer 2009 4. Christopher Bishop, Pattern Recognition and Machine Learning, Springer, 2007.				
Learning Outcomes				
- Conceptual Mastery: Students will understand key concepts and techniques in machine learning, including different types of learning and model evaluation metrics. - Algorithm Application: Students will be able to select and apply suitable machine learning algorithms to solve specific problems. - Implementation Skills: Students will be proficient in using tools such as Python, scikit-learn, TensorFlow, or PyTorch to build and evaluate machine learning models. - Data Proficiency: Students will demonstrate the ability to preprocess and visualize data, and perform feature selection and extraction. - Ethical Awareness: Students will recognize and address ethical issues and biases in machine learning, ensuring responsible and fair use of technology.				

CSCMJ702	Research Methodology	Theory	Practical	Cumulative
Major		2	0	2
		30	0	30
Course Objectives				
- To study and understand the research goals, research issues & challenges, scientific methods. - Reviewing Literature and Research Papers; Writing Research Papers, Thesis, Reports and Project Proposals Plagiarism and Copyrights.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Research Foundations: Research: Meaning, Characteristics, Objectives, Types, Variables, Steps of research, Hypothesis, Data, Interdisciplinary research, Tran disciplinary research and Multidisciplinary research. Literature: Primary, Secondary and Tertiary sources, Problem identification, Scholarly databases, Types of research papers. Research Metrics: Journal level Metrics and Author level metrics, Publication Identifiers, Abstracting Databases, and Publishers. Terminology: Conference, Symposium, Congress, Seminar, Workshop, Colloquium, Patents.			8
II	Research and Publication Ethics: Introduction to Philosophy and Ethics; Ethics with respect to science and research, intellectual honesty, and research integrity; Falsification, Fabrication, and Plagiarism (FFP); plagiarism types, use of plagiarism software and open-source software tools; Uses and misuses of AI tools in research and its boundary;; Copyrights; duplicate and redundant publications, salami slicing; selective reporting and misrepresentation of data; Definition, introduction, and importance of publication ethics; conflict of interest; Publication misconduct definition, problems that lead to unethical behavior and vice-versa, types; violation of publication ethics, authorship, and contributor ship; identification of publication misconduct, complaints, and appeals; predatory publishers and journals; subject-specific ethical issues, FFP and authorship; Complaints and appeals regarding fraud from India and abroad.			10
III	Tools and Applications. Meaning and Importance of Data, Sources of Data, Use of Secondary Data, Data Collection Techniques, Observation Method, Experimentation, Simulation. Types of Data, Measurement of Scales and Indices, Pilot Studies and Pre-tests.			6
IV	Technical Paper Writing: Writing Research Papers, Synopsis, Thesis, Reports and Project Proposals; Formatting, Appendices, Citation Formats and Style; General Conventions, Issues, Plagiarism and Copyrights. Reference Managers. Format and Style of Referencing.			6
Texts / References Books				
- Kothari, C.R., 2004. Research methodology: Methods and techniques. New Age International. - Cooper, D.R., Schindler, P.S. and Sun, J., 2006. Business research methods (Vol. 9). New York: McGraw-Hill Irwin. - Creswell, J.W. and Creswell, J.D., 2017. Research design: Qualitative, quantitative, and mixed methods approaches. Sage publications. - Krishnaswamy, K.N., 2006. Management Research Methodology: Integration of Principles, Methods and Techniques. Pearson Education India. - Sekaran, U. and Bougie, R., 2016. Research methods for business: A skill building approach. John Wiley & Sons. - Advanced Research Methodology by R. Barker Bausell. Scarecrow Press 2013 - Social Sciences by Bill Taylor, Gautam Sinha, Taposh Ghoshal. Prentice-Hall of India Private Limited, New Delhi. 2006				
Learning Outcomes				
After successful completion of this course, students will be able to:				
- Understand the basic concepts of research and its methodologies, Identify appropriate research topics, select and define appropriate research problem and parameters - Prepare a project proposal (to undertake a project) - Organize and conduct research in a more appropriate manner, writing research reports and thesis.				

CSCMJ703	Operations Research	Theory	Practical	Cumulative
Major		3	1	3
		45	30	75
Course Objectives				
• Optimization Techniques Mastery: Introduce students to fundamental optimization techniques and methods used in operations research. • Decision-Making Skills: Enhance students' abilities to make informed and effective decisions using quantitative analysis. • Modeling Proficiency: Develop skills in formulating, analyzing, and solving linear and nonlinear models. • Application of Tools: Provide practical experience with operations research tools and software for real-world problem-solving. • Interdisciplinary Approach: Encourage an interdisciplinary approach to solving complex problems in various fields, including logistics, manufacturing, and service industries.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Introduction to Theory of Optimization & Linear Programming: Features of OR, Modeling in Operations Research, Classification of Models, General Solution Methods for O.R Models, Scientific Method in OR, Methodology of OR, Applications, Opportunities and Shortcomings of O.R. Applications of operation research in economics, Finance and decision making, Formulation of Mathematical Model of Linear Programming Problem.			10
II	Graphical Solution and Simplex Method Graphical Solution: Solution of Linear Programming Problem by Graphical Method, Special Cases: (i) Alternate Optima (ii) Unbounded Solution (iii) Infeasible Solution Simplex Method: Introduction to Simplex Method- Maximization and Minimization, Duality, primal to dual conversion only.			10
III	Transportation Models & Assignment Models: Balanced and Unbalanced Models of Transportation, Initial Basic Feasible Solutions (i). North-West Corner Method, (ii) Row Minima Method, (iii) Column Minima Method (iv) Matrix Minima Method, (v) Vogel’s Approximation Method, Modi Method, Degeneracy. Balanced and Unbalanced Assignment Models, Hungarian Method, Maximization and Minimization.			15
IV	Game Theory & Queuing Theory: Game Theory: Two-Person Zero sum Games, Some Basic Terms, The Maximin-Minimax Principle, Games without Saddle Points (Mixed Strategies), Graphical solution of $2 \times n$ and $m \times 2$ Games, algebraic method. Queuing Theory: Introduction; Basic Definitions and Notations; Axiomatic Derivation of the Arrival & Departure (Poisson Queue). Pure Birth and Death Models; Poisson Queue Models: $M/M/1 : \infty /FIFO$ and $M/M/1: N/ FIFO$. Network Analysis: Shortest Path; Dijkstra Algorithm; Floyd Algorithm; Maximal Flow Problem (Ford-Fulkerson); PERT-CPM (Cost Analysis, Crashing, Resource Allocation excluded).			10
Texts / References Books				
1. Swarup K., P.K. Gupta, and Manmohan, Operations Research, Sultan Chand: New Delhi, 12 th thoroughly revised Ed. 2. Taha, H. A. (2013). Operations research: an introduction. Pearson Education India. 3. Hillier, F. S., & Lieberman, G. J. (2015). Introduction to operations research. McGraw-Hill. 4. J.K. Sharma, Operations Research Theory; Application, Macmillan, 3rd Ed. 5. Hillier, F. S., & Lieberman, G. J. (2015). Introduction to operations research. McGraw-Hill.				
Learning Outcomes				
• Conceptual Understanding: Students will understand the basic concepts and techniques of operations research, including linear programming, integer programming, and network models. • Analytical Skills: Students will be able to analyze complex problems and develop appropriate mathematical models. • Solution Techniques: Students will learn and apply various optimization techniques to find the best possible solutions. • Software Proficiency: Students will gain hands-on experience with operations research software and tools. • Practical Application: Students will demonstrate the ability to apply operations research methods to real-world scenarios, improving decision-making and operational efficiency.				

CSCMJ704	Cloud Computing	Theory	Practical	Cumulative
Major		3	1	4
		45	30	75
Course Objectives				
- To study and understand the fundamentals and essentials of Cloud Computing. - To provide a sound foundation of the Cloud Computing so that they are able to start using and adopting Cloud Computing services and tools in their real-life scenarios. - To enable students exploring some important cloud computing driven commercial systems and applications. - To expose the students to frontier areas of Cloud Computing and information systems, while providing sufficient foundations to enable further study and research.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Introduction to Parallel and Distributed Computing; Introduction to Cloud Computing; Characteristics and benefits of cloud computing; Historical developments and evolution of cloud computing: Distributed Systems, Virtualization, Web 2.0, Service-oriented computing, Utility Computing; Cloud Computing Reference Model.			12
II	Introduction to virtualization; Characteristics of virtualized environments; Taxonomy of virtualization techniques; Virtualization and cloud computing; Pros and cons of virtualization; Technology examples: Xen: paravirtualization, VMware: full virtualization, Microsoft Hyper-V.			9
III	Cloud Computing Architecture; Service models: Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS); Deployment models: Public, Private, Hybrid, Community; IaaS: Introduction to IaaS, Resource Virtualization i.e. Server, Storage and Network virtualization; PaaS: Introduction to PaaS, Cloud platform & Management of Computation and Storage; SaaS: Introduction to SaaS, Cloud Services, Web services, Web 2.0, Web OS; Case studies related to IaaS, PaaS and SaaS.			12
IV	Economics of the cloud; Open Challenges in Cloud Computing; Introduction to emerging computing paradigms and research challenges: Edge Computing, Mobile Cloud Computing, Fog Computing etc.; Introduction to IoT Cloud; Study on simulators related to cloud computing and emerging computing paradigms.			12
Texts / References Books				
1. R. Buyya, C. Vecchiola, S. ThamaraiSelvi, Mastering Cloud Computing, McGraw Hill Education. 2. B.Sosinsky, Cloud Computing Bible, Wiley. 3. K. Hwang, G. C. Fox,J.Dongarra, Distributed and Cloud Computing: From Parallel Processing to the Internet of Things, Morgan Kaufmann				
Learning Outcomes				
After successful completion of this course, student will be able to - Explain the core concepts of the cloud computing paradigm: how and why this paradigm shift came about, the characteristics, advantages and challenges brought about by the various models and services in cloud computing. - Apply the fundamental concepts in datacenters to understand the tradeoffs in power, efficiency and cost. - Identify resource management fundamentals, i.e. resource abstraction, sharing and sandboxing and outline their role in managing infrastructure in cloud computing. - Analyze various cloud programming models and apply them to solve problems on the cloud.				

CSCMJ705	Soft Computing	Theory	Practical	Cumulative
Major		3	1	4
		45	30	75
Course Objectives				
● Introduction to Soft Computing: Provide a comprehensive understanding of soft computing techniques and their applications. ● Conceptual Framework: Develop a solid foundation in the theoretical concepts of fuzzy logic, neural networks, and evolutionary algorithms. ● Problem-Solving Skills: Enhance students' abilities to apply soft computing methods to solve complex real-world problems. ● Interdisciplinary Knowledge: Foster an understanding of the interdisciplinary nature of soft computing and its integration with other fields. ● Practical Applications: Equip students with practical skills to implement and experiment with soft computing techniques in various domains.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Introduction to Soft Computing:Concept of computing systems, Soft computing versus Hard computing, Characteristics of Soft computing, Some applications of Soft computing techniques			10
II	An Introduction to Artificial Neural Network: Fundamental concepts, Evolution of NN, BasicModels of ANN, connections and learning, Terminologies such as weights, Bias, Threshold,Learning Rate etc., McCulloch-Pitts Neuron, Heb Network; Supervised Learning Network: Perceptron Network, Adaptive Linear Neuron, Multiple Adaptive Linear Neurons, BackPropagation Network, Radial Basis Function Network; Associate Memory Networks:Introduction and training algorithm for pattern association, Auto-associative Memory Network, Hetro-associative Memory Network, Bidirectional associative memory, Hopfield Network;Unsupervised Learning Network: Introduction; Fixed Weight Competitive Nets; Kohonen Self-Organizing Feature Maps; Adaptive Resonance Theory; Applications of ANN: Applications:such as recognition of characters, fabric defect identification etc.			10
III	Introduction to Fuzzy Logic: Classical Sets, Fuzzy Sets: operations and properties. Operations on fuzzy relations; Membership functions: Features, fuzzification, methods of membership value assignments; Defuzzification: Introduction; Lambda-Cuts for fuzzy sets and fuzzy relations; Defuzzification methods; Fuzzy Rules: Introduction; formation of rules, decomposition and aggregation of rules; fuzzy reasoning; Fuzzy inference systems (FIS) and applications: FIS methods: Mamdani and Sugeno; Applications: such as fuzzy logic control etc.			15
IV	Genetic Algorithms: Concept of Genetics and Evolution and its application to probabilistic search techniques, Basic GA framework and different GA architectures, GA operators: Encoding, Crossover, Selection, Mutation, etc., Solving single objective optimization problems using GAs.			10
Texts / References Books				
1. Timothy J. Ross, "Fuzzy Logic with Engineering Applications," McGraw Hill, 1995. 2. Simon Haykin, "Neural Networks" Pearson Education. 3. B. Yegnanarayana, "Artificial Neural Networks," PHI, India, 2006. 4. S. N. Sivanandan and S. N. Deepa, "Principles of Soft Computing", Wiley India, 2012. 5. Fakhreddine O. Karray and Clarence De Silva., "Soft Computing and Intelligent SystemsDesign, Theory, Tools and Applications," Pearson Education, India, 2009.				
Learning Outcomes				
● Conceptual Clarity: Students will understand the fundamental principles of soft computing techniques such as fuzzy logic, neural networks, and genetic algorithms. ● Analytical Skills: Students will be able to analyze and design solutions for complex problems using soft computing methods. ● Implementation Skills: Students will gain proficiency in implementing soft computing algorithms and models using appropriate software tools. ● Application Insight: Students will demonstrate the ability to apply soft computing techniques to real-world problems in areas like optimization, control systems, and pattern recognition. ● Interdisciplinary Approach: Students will appreciate the interdisciplinary nature of soft computing and its potential to innovate across various fields.				

CSCMJ706	Network Security and Cryptography	Theory	Practical	Cumulative
		3	1	4
Major		45	30	75
Course Objectives				
• To understand basics of Cryptography and Network Security. • To make the student learn different encryption techniques along with hashfunctions, MAC, digital signatures and their use in various protocols for network security and system security. • To learn about how to maintain the Confidentiality, Integrity and Availability of data. • To understand various protocols for network security to protect against the threats in the networks.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Introduction: Computer Security Concepts, The OSI Security Architecture, Security Attacks, Security Services, Security Mechanisms, A Model for Network Security. Symmetric Encryption and Message Confidentiality: Symmetric Encryption Principles, Traditional Ciphers, Stream Ciphers and Block Ciphers, Modern Symmetric Block Ciphers: DES, AES, Cipher Block Modes of Operation.			15
II	Public-Key Cryptography and Message Authentication: Approaches to Message Authentication, Secure Hash Functions, Message Authentication Codes, Public-Key Cryptography Principles, Public-Key Cryptography Algorithms: RSA, Digital Signatures.			10
III	Key Distribution and User Authentication: Symmetric Key Distribution Using Symmetric Encryption, Kerberos, Key Distribution Using Asymmetric Encryption, X.509 Certificates. Transport-Level Security: Web Security Considerations, Secure Socket Layer and Transport Layer Security, HTTPS, Secure Shell (SSH).			10
IV	IP Security: Overview of IP Security (IPSec), IP Security Architecture, Modes of Operation, Security Associations (SA), Authentication Header (AH), Encapsulating Security Payload (ESP), Internet Key Exchange. Electronic Mail Security: Threats to E-Mail, Requirements and Solutions, Encryption for Secure E-Mail, Secure E-Mail System.			10
Texts / References Books				
1. Stalling, W., Network security essentials, Pearson education Asia publication. 2. Kahate, A., Cryptography & Network Security, Mc Graw Hill. 3. Forouzan, B. A., Cryptography and Network Security, McGraw-Hill Education.				
Learning Outcomes				
After successful completion of this course, student will be able to:				
• Analyze and design classical encryption techniques and block ciphers. • Understand and analyze data encryption standard. • Understand and analyze public-key cryptography, RSA and other public-key cryptosystems such as Diffie-Hellman Key Exchange etc. • Understand key management and distribution schemes and design User Authentication Protocols. • Analyze and design hash and MAC algorithms, and digital signatures. • Design network application security schemes, such as PGP, S/MIME, IPSec, SSL, TLS				

CSCMJ707	Heuristics & Metaheuristics	Theory	Practical	Cumulative
		3	1	4
Major		45	30	75
Course Objectives				
• Problem-Solving Techniques: Introduce students to heuristic and metaheuristic techniques for solving complex optimization problems. • Algorithm Understanding: Provide a thorough understanding of various heuristic algorithms, such as greedy algorithms, local search, and evolutionary algorithms. • Design and Implementation: Equip students with the skills to design, implement, and evaluate heuristic and metaheuristic algorithms. • Application Knowledge: Demonstrate how these techniques can be applied to real-world problems in various domains, such as logistics, scheduling, and machine learning. • Comparative Analysis: Teach students how to compare and contrast different heuristic and metaheuristic methods in terms of efficiency, accuracy, and computational complexity.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Introduction to Computability Theory: NP-Complete problems; Landscape and Search Space: topologies, meaning and importance; Traditional vs Heuristic Search; Greedy algorithms and Exact Methods; Single solution Metaheuristics; Review of Single State Methods: Hill Climbing, Tabu Search, Iterated Local Search. Models for optimization: (i) Single-Objective Optimization; (ii) Constrained Optimization; (iii) Multi-Objective Optimization; (iv) Optimization under Uncertainty; (v) Dynamic Optimization.			10
II	Bio-Inspired Algorithms: (i) Common concepts; (a) Selection methods; (b) Reproduction; (c) Replacement strategies; (ii) Genetic Algorithmss; (ii) Genetic Programmings (iii) AIS; (iv) Swarm Intelligence: Particle Swarm Optimization, Ant Colony Optimization, Artificial Bee Colony; (v) Differential Evolution ; and (vi) Estimation of Distribution Algorithms.			10
III	Hybrid Metaheuristics; Parallel Metaheuristics; Multi Objective Optimization by Metaheuristics; Machine Learning; Metaheuristics.			15
IV	Metaheuristics application examples in: Network Sciences; Games; Internet of Things; Computer Security; Robotics; Art and Design.			10
Texts / References Books				
1. Larrañaga, P., & Lozano, J. A. (Eds.). (2001). Estimation of distribution algorithms: A new tool for evolutionary computation (Vol. 2). Springer Science & Business Media. 2. Kearns, M. J., & Vazirani, U. (1994). An introduction to computational learning theory. MIT press. 3. S. Luke, Essentials of Metaheuristics, 2015. (http://cs.gmu.edu/~sean/book/metaheuristics/). 4. Deb, K. (2001). Multi-objective optimization using evolutionary algorithms (Vol. 16). John Wiley & Sons. 5. Talbi, E. G. (2009). Metaheuristics: from design to implementation. John Wiley & Sons.				
Learning Outcomes				
• Algorithm Proficiency: Students will gain proficiency in various heuristic and metaheuristic algorithms. • Problem-Solving Skills: Students will be able to apply these techniques to solve complex optimization problems effectively. • Implementation Skills: Students will demonstrate the ability to design, implement, and test heuristic and metaheuristic algorithms using programming languages or software tools. • Evaluation and Analysis: Students will be able to evaluate the performance of different algorithms and select the most appropriate one for a given problem. • Practical Application: Students will understand how to apply heuristics and metaheuristics to practical problems in diverse fields, enhancing their problem-solving toolkit.				

CSCMJ708	Knowledge & Data Discovery	Theory	Practical	Cumulative
		3	1	4
Major		45	30	75
Course Objectives				
• Ability to understand the role of data mining in knowledge discovery process. • To familiarize with various data mining functionalities and how it can be applied to various real-world problems. • To learn about finding data characteristics and evaluating the outcome of data mining process. • To familiarize with various machine learning algorithms used in data mining.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Basics of Knowledge and Data Discovery (KDD) Tasks: Classification, Regression, Time series analysis, prediction, clustering, summarization, association rules, sequence discovery. KDD vs. data mining, data discovery issues, data discovery metrics, data discovery vs data mining vs database.			10
II	Related concepts: database/ OLTP systems, information retrieval, decision support system, dimensional modeling, data warehousing, OLAP, web search engines, statistics, machine learning, pattern matching. Statistical perspective on data discovery: point estimation, models based on summarization, Bayes theorem, hypothesis testing, regression and correlation. Similarity measures, decision trees, neural networks, genetic algorithms.			10
III	Classification algorithm instatistical, distance, tree neural network, and rule framework. Hierarchical, and partitionbased clustering algorithm.Algorithm for clustering large databases, clustering with categorical attributes, association rules.			10
IV	Web content , structure , and web mining. Spatial data overview, spatial data primitives, generalization and specialization, spatial association rules, and spatial classification and clustering algorithms. Temporal data overview, modelling of temporal events, time series data analysis, transformation, and prediction. Pattern detection, sequence analysis algorithm (AprioriAll, SPADE, Generalization, Feature Extraction), temporal association rules.			15
Texts / References Books				
1. J. Han and M. Kamber, “Data Mining: Concepts and Techniques”, Academic Press. 2. I. H. Witten et al., Data Mining: Practical machine Learning Tools and Techniques, Morgan Kaufmann Publisher. 3. Pang-Ning Tan, Michael Steinbach, Vipin Kumar, Introduction to Data Mining, Pearson Education, 2016 4. Margaret H. Dunham, Data Mining: Introductory and Advanced Topics, Pearson, 2002. 5. Ian H. Witten, Eibe Frank, Mark A. Hall, Christopher J. Pal, Morgan Kaufmann, 2016.				
Learning Outcomes				
After successful completion of this course, student will be able to • Apply data mining functionalities on real world problems and datasets. • Have some knowledge about the couple of data mining tools and how they can be used for large data. • Find the characteristics of given data and may identify presence of outliers, if any. • Pursue some advance course on data science and may help in subjects like Big Data, AI etc.				

CSCMJ709	Distributed Systems	Theory	Practical	Cumulative
		3	1	4
Major		45	30	75
Course Objectives				
- To learn core concepts such as distributed computing, concurrency, fault tolerance, and scalability to design and manage distributed architectures effectively. - To apply principles to develop, deploy, and analyze distributed systems, including communication protocols, data consistency models, and performance optimization techniques.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Introduction: Definition, Goals, Types of Distributed Systems, Distributed Computing Systems, Distributed Information Systems, Distributed Pervasive Systems, System Architectures, Architectures versus Middleware, Self-Management in Distributed. Processes & Communication: Threads, Threads in Distributed Systems, Virtualization, Clients, Servers, Code Migration, Remote Procedure Call, Message-Oriented Communication, Stream-Oriented Communication, Multicast Communication.			13
II	Naming: Names, Identifiers, Addresses, Flat Naming, Structured Naming, Attribute-Based Naming. Synchronization: Clock Synchronization, Logical Clocks, Mutual Exclusion, Centralized Algorithm, Decentralized Algorithm, Distributed Algorithm, Token Ring Algorithm, Global Positioning of Nodes, Election Algorithms.			12
III	Consistency, Replication & Fault Tolerance: Introduction, Reasons for Replication, Replication as Scaling Technique, Data-Centric Consistency Models, Client-Centric Consistency Models, Replica Management, Consistency Protocols, Process Resilience, Reliable Client-Server Communication, Reliable Group Communication, Distributed Commit, Recovery.			10
IV	Distributed Shared Memory & File Systems: Hardware DSM, Design issues in DSM Systems, Implementation issues, Heterogeneous and Other DSM Systems, Distributed File Systems, File Service Architecture, Case Study: Sun Network File System, The Andrew File System.			10
Texts / References Books				
A. S. Tanenbaum, M. V. Steen, Distributed systems: principles and paradigms, Prentice-Hall. G. F. Coulouris, J. Dollimore, T. Kindberg, Distributed Systems: Concepts and Design, Addison-Wesley. M. Singhal, N. G. Shivaratri, Advanced concepts in operating systems, McGraw-Hill.				
Learning Outcomes				
After successful completion of this course, student will be able to:				
- Develop the ability to design scalable and fault-tolerant distributed systems, applying concepts such as consensus algorithms, replication, and partitioning. - Gain hands-on experience in implementing distributed systems, including network communication, data consistency, and distributed algorithms. - Evaluate the performance and scalability of distributed systems, using metrics and tools to identify bottlenecks and optimize efficiency. - Develop skills to troubleshoot issues and maintain distributed systems, ensuring reliability, fault tolerance, and effective error handling.				

CSCMJ710	Parallel Computing	Theory	Practical	Cumulative
Major		3	1	4
		45	30	75
Course Objectives				
- To provide the Understanding of hardware and software work together in parallel and sequential mode. - To understand the simple systems to complex shared resource architectures and learning parallel programming techniques. - To discuss and understand the common software packages and libraries, learning how to write and document software packages, enhancing skills in managing simultaneous data processes, optimizing computational tasks for quicker execution. - To understand the fundamental aspects of parallel systems, and providing an integrated view of the various facets of software development on parallel systems.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Introduction to Parallel Computing: Supercomputers and grand challenge problems, Modern Parallel Computers, Data Dependence Graph, Data Parallelism, Functional Parallelism, Pipelining and Data Clustering.			10
II	Interconnection Networks: Switch Network Topologies, Direct and Indirect Network Topology, Bus, Star, Ring, Mesh, Tree, Binary Tree Network, Hyper Tree Network, Hybrid, Hypercube, Perfect Shuffle Network, Torus and Butterfly Network.			10
III	Performance Analysis: Introduction, Execution Time, Speedup, Linear and Superliner Speedup, Efficacy and Efficiency, Amdahl's Law and Amdahl Effect, Gustafson-Barsis's Law, Minsky's Conjecture, The Karp-Flatt Metric, The Isoefficiency Metric, Isoefficiency Relation, Cost and Scalability.			10
IV	Parallel Computational Models: Flynn's Taxonomy, PRAM, EREW, CREW, ERCW, CRCW, Simulating CRCW, CREW & EREW, PRAM algorithms. Introduction to Parallel Algorithms: Parallel Programming Models, PVM, MPI Paradigms, Parallel Programming Language, Brent's Theorem, Simple parallel programs in MPI environments, Parallel algorithms on network, Addition of Matrices, Multiplication of Matrices.			15
Texts / References Books				
- Hwang and Briggs, Computer Architecture and Parallel Processing, McGraw Hill. - Crichlow, Introduction to Distributed and Parallel Computing, PHI. - M.J. Quinn, Designing Efficient Algorithms for Parallel Computers, McGraw-Hill. V. Rajaraman, Elements of Parallel Computing, PHI. - Joseph JA JA, Introduction to Parallel Algorithms, Addison Wesley.				
Learning Outcomes				
After successful completion of this course, students will be able to: - Describe different parallel architectures, inter-connect networks, programming models and develop an efficient parallel algorithm to solve given problem. - Analyze and measure performance of modern parallel computing systems. - Build the logic to parallelize the programming task. - Develop efficient parallel algorithms for sorting and Graph. - Develop an algorithm for CUDA Architecture.				

CSCMJ711	Image Processing	Theory	Practical	Cumulative
Major/Minor		3	1	4
		45	30	75
Course Objectives				
- To learn digital image processing fundamentals, including visual perception, image formation, and pixel operations. - To develop skills in image enhancement through techniques like contrast stretching, histogram processing, and spatial and frequency domain filtering. - To study image reconstruction methods from projections, such as tomography and Radon transforms, with a focus on Fourier reconstruction and fan-beam reconstruction, essential for medical imaging and diagnostics. - To explore image compression principles, including error criteria, lossy methods like JPEG, and lossless techniques such as Huffman coding, critical for efficient data storage and transmission in the era of big data and multimedia.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Digital Image Fundamental: Elements of Visual Perception- Structure of the human eye, Image formation in the eye, brightness adaptation and discrimination; light and electromagnetic spectrum, image sensing and acquisition, image sampling and quantization, basic relationships between pixels, linear and nonlinear operations.			10
II	Image Enhancement: Point processing: Contrast stretching, power-law and gamma transformation. Histogram processing: histogram equalization and matching. Filtering and Restoration: Degradation function and Noise Models, Spatial Domain Filtering: Correlation and Convolution, Smoothing Linear and Nonlinear Filters, Sharpening Linear and Nonlinear Filters. Frequency Domain Filtering: Filtering: Low-pass (Smoothing) & High-Pass (Sharpening), Homomorphic Filtering, Periodic Noise Reduction and Inverse Filtering & Wiener Filtering.			15
III	Image Reconstruction from Projections: Transmission tomography, reflection tomography, emission tomography, magnetic resonance imaging, and projection based image processing. Radon transform, back projection operator, projection theorem, inverse radon transform, convolution filter back projection, reconstruction from blurred noisy projections, Fourier reconstruction, fan-beam reconstruction.			10
IV	Image Compression: Introduction, Error criterion- objective and subjective criterion, Lossy compression- transform domain compression, JPEG compression, block truncation compression, vector quantization compression; Lossless compression- Huffman coding, arithmetic coding, transformed coding, run-length coding..			10
Texts / References Books				
A. K. Jain, Fundamentals of Digital Image Processing, Pearson Education India, 2015. - Rafael Gonzalez, Richard Woods, Digital Image Processing, Pearson Education India, 2017				
Learning Outcomes				
After successful completion of this course, students will be able to:				
- Master digital image processing fundamentals, including visual perception and pixel operations. - Enhance image quality using techniques such as contrast stretching and histogram processing. - Gain proficiency in spatial and frequency domain filtering techniques. - Understand and implement advanced image reconstruction methods from projections, including tomography and Radon transforms. - Apply image compression methods, including both lossy techniques (e.g., JPEG) and lossless techniques (e.g., Huffman coding), for optimal data storage and transmission. - Address contemporary challenges in medical imaging, multimedia, and data analysis with these skills.				

Semester- VIII

CSCMJ801	Deep Learning	Theory	Practical	Cumulative
		3	1	4
Major/Minor		45	30	75
Course Objectives				
- Core Principles: Introduce students to the fundamental concepts and theories of deep learning, including neural networks and their architectures. - Hands-On Experience: Develop practical skills through hands-on experience with deep learning frameworks and tools. - Model Building: Teach students how to design, train, and evaluate deep learning models for various applications. - Advanced Techniques: Explore advanced topics such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and generative models. - Real-World Applications: Provide an understanding of how to apply deep learning techniques to real-world problems in areas such as computer vision, natural language processing, and robotics.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Linear Algebra Review: Brief review of concepts from Linear Algebra. Optimization: Types of errors, bias-variance tradeoff, overfitting, underfitting, a brief review of concepts from Vector Calculus and optimization, variants of gradient descent, momentum. Logistic Regression: Basic concepts of regression and classification problems, linear models addressing regression and classification, maximum likelihood, logistic regression classifiers.			10
II	Neural Networks: Basic concepts of artificial neurons, single and multilayer perceptrons, perceptron learning algorithm, its convergence proof, different activation functions, softmax crossentropy loss function.			10
III	Recurrent Neural Networks: Discussion on Recurrent Neural Networks (RNNs), Long-Short, Term Memory (LSTM) architectures, and basics of word embedding. ConvNets: Basic concepts of Convolutional Neural Networks starting from filetering. Convolution and pooling operation and arithmetics of these. ConvNet Architectures: Discussions on famous convnet architectures AlexNet, ZFNet, VGG, C3D, GoogLeNet, ResNet, MobileNetv1.			15
IV	Introduction to Deep Reinforcement Learning, Autoencoders (standard, denoising, contractive, etc).			10
Texts / References Books				
- Goodfellow, I., Bengio, Y., Courville, A. (2016). Deep learning. MIT press. (http://www.deeplearningbook.org) - Nielsen, M. A. (2015). Neural networks and deep learning (Vol. 25, pp. 15-24). San Francisco, CA, USA: Determination press. - Chollet, F. (2021). Deep learning with Python. Simon and Schuster. - Géron, A. (2022). Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow. Reilly Media, Inc. - Jones H. (2018). Deep Learning: An Essential Guide to Deep Learning for Beginners Who Want to Understand How Deep Neural Networks Work and Related to Machine Learning and Artificial Intelligence. Create space Independent Publishing				
Learning Outcomes				
- Conceptual Mastery: Students will understand the key principles and theories underlying deep learning and neural networks. - Practical Skills: Students will gain proficiency in using deep learning frameworks like TensorFlow and PyTorch to build models. - Model Development: Students will be able to design, implement, and evaluate deep learning models for various tasks. - Advanced Knowledge: Students will have a deep understanding of advanced deep learning techniques, including CNNs, RNNs, and GANs. - Application Ability: Students will be capable of applying deep learning methods to solve complex real-world problems in various domains.				

CSCMJ802	Internet of Things	Theory	Practical	Cumulative
		3	1	4
Major		45	30	75
Course Objectives				
- To understand the application areas of IOT. - To realize the revolution of Internet in Mobile Devices, Cloud & Sensor Networks. - To understand building blocks of Internet of Things and characteristics.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Introduction to IoT: Genesis of IoT, IoT and Digitization, IoT Challenges, Comparing IoT architectures, a simplified IoT architecture, The core IoT functional Stack, IoT data management and compute stack.			11
II	Engineering for IoT Networks: Sensors, Actuators, Smart Objects, Sensor Networks, IoT Access Technologies, IP as the IoT Network Layer, Applications protocols for IoT.			12
III	Data and Analytics for IoT: An introduction to data analytics for IoT, Machine Learning, Big data analytics tools and technology, edge streaming analytics, network analytics.			12
IV	IoT in Industry: Manufacturing, Oil and Gas, Utilities, Smart and Connected Cities, Transportation, Mining, Public Safety.			10
Texts / References Books				
D. Hanes, G.Salgueiro, P.Grossetete, R. Barton, J. Henry, IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things, CISCO. - Rajkamal, Internet of Things, McGraw Hill Education.				
Learning Outcomes				
After successful completion of this course, students will be able to: - Identify the Components that form part of IoT Architecture. - Determine the most appropriate IoT Devices and Sensors based on Case Studies. - Setup the connections between the Devices and Sensors. - Evaluate the appropriate protocol for communication between IoT and to analyze the communication protocols for IoT.				

CSCMJ803	Blockchain Technology	Theory	Practical	Cumulative
		3	1	4
		45	30	75
Major				
Course Objectives				
- The course aims to provide a comprehensive understanding of blockchain technology, and its applications. - Students will first explore the fundamentals of distributed databases, fault tolerance, and cryptographic techniques including hash functions, digital signatures, and zero-knowledge proofs. - Focus will be on blockchain technology, covering its structure, consensus mechanisms (including Nakamoto consensus and variants), mining strategies, smart contracts, and vulnerabilities. - Cryptocurrencies like Bitcoin and Ethereum will be studied in detail, along with their protocols and applications.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Basics: Distributed Database, Two General Problem, Byzantine General problem and Fault Tolerance, Hadoop Distributed File System, Distributed Hash Table, ASIC resistance, Turing Complete. Cryptography: Hash function, Digital Signature - ECDSA, Memory Hard Algorithm, Zero Knowledge Proof.			10
II	Blockchain: Introduction, Advantage over the conventional distributed database, Blockchain Network, Mining Mechanism, Distributed Consensus, Merkle Patricia Tree, Gas Limit, Transactions and Fee, Anonymity, Reward, Chain Policy, Life of Blockchain application, Soft and Hard Fork, Private and Public blockchain.			10
III	Distributed Consensus: Nakamoto consensus, Proof of Work, Proof of Stake, Proof of Burn, Difficulty Level, Sybil Attack, Energy Utilization, PBFT and its variants.			10
IV	Cryptocurrency: History, Distributed Ledger, Bitcoin protocols - Mining strategy and rewards, Ethereum - Construction, DAO, Smart Contract, GHOST, Vulnerability, Attacks, Sidechain, Namecoin. Blockchain Applications: Internet of Things, Medical Record Management System, Domain Name Service and future of Blockchain.			15
Texts / References Books				
- Narayanan, A., Bonneau, J., Felten, E., Miller, A., & Goldfeder, S. (2016). Bitcoin and cryptocurrency technologies: a comprehensive introduction. Princeton University Press. - Wattenhofer, R. (2016). The science of the blockchain. CreateSpace Independent Publishing Platform. - Antonopoulos, A. M. (2017). Mastering Bitcoin: Programming the open blockchain. O'Reilly Media, Inc. - Saurabh, K., & Saxena, A. (2020) Blockchain Technology: Concepts and Applications. Wiley.				
Learning Outcomes				
- This course equips students with a thorough understanding cryptography, and blockchain technology. - The focus on blockchain covers its architecture, consensus mechanisms, mining strategies, and smart contract applications. Students will evaluate real-world applications of blockchain in sectors like IoT, healthcare, and finance, assessing both advantages and challenges. - The course fosters critical thinking skills for analyzing blockchain protocols, identifying vulnerabilities, and proposing enhancements in security and efficiency. - By the end of the course, students will be prepared for roles in blockchain development, cybersecurity, and other fields leveraging decentralized technologies.				

CSCMJ804	Information Retrieval	Theory	Practical	Cumulative
		3	1	4
Major		45	30	75
Course Objectives				
- To provide an overview of Information Retrieval. - To introduce students about insights of the several topics of Information retrieval such as – Boolean retrieval model, Vector space model, Latent semantic indexing, XML and Image retrieval model. - To provide comprehensive details about various Evaluation methods. - To provide implementational insight about the topics covered in the course.				
Syllabus				
Units	Course Content	Hr. of Teaching		
I	Introduction Information Retrieval: Information retrieval process, Indexing, Information retrieval model, Boolean retrieval model. Dictionary and Postings: Tokenization, Stop words, Stemming, Inverted index, Skip pointers, Phrase queries.	10		
II	Tolerant Retrieval: Wild card queries, Permuterm index, Bigram index, Spelling correction, Edit distance, Jaccard coefficient, Soundex. Term Weighting and Vector Space Model: Wild card queries, Permuterm index, Bigram index, Spelling correction, Edit distance, JaccardJaccard coefficient, Soundex.	10		
III	Evaluation: Precision, Recall, F-measure, E-measure, Normalized recall, Evaluation problems. Latent Semantic Indexing: Eigen vectors, Singular value decomposition, Lowrank approximation, Problems with Lexical Semantics.Query Expansion: Relevance feedback, Rocchio algorithm, Probabilistic relevance feedback, Query Expansion and its types, Query drift. Probabilistic Information Retrieval: Probabilistic relevance feedback, Probability ranking principle, Binary Independence Model, Bayesian network for text retrieval.	13		
IV	XML Indexing and Search: Data vs. Text-centric XML, Text-Centric XML retrieval, Structural terms. Content BasedImage Retrieval: Introduction to content Based Image retrieval, Challenges in Image retrieval, Image representation, Indexing and retrieving images, Relevance feedback. Introduction to video retrieval.	12		
Texts / References Books				
C. D. Manning, P. Raghvan, H. Schutze, Introduction to Information Retrieval, Cambridge University Press, 2008. R. Mihalcea, D. Radev, Graph based Natural Language Processing and Information Retrieval, Cambridge University Press, April 2011. - U.S. Tiwary, T. Siddiqui, Natural Language Processing and Information Retrieval, Oxford University Press, 2008. G. S. Ingersol, T. S. Morton, A. L. Farris, Taming Text: How to Find, Organize and Manipulate It, Manning Publications, 2013. S. Bird, E. Klein, E. Loper, Natural Language Processing with Python, O'Reilly Media, 2009.				
Learning Outcomes				
- Students will get the understanding different Information retrieval model. - Students will get to know about evaluation methods of the information retrieval model. - Students will get to know the challenges associated with each topic.				

CSCMJ805	Statistical Pattern Recognition	Theory	Practical	Cumulative
		3	1	4
Major		45	30	75
Course Objectives				
- To gain a profound understanding of the theories, algorithms, and applications of the state-of-the-art of statistical pattern recognition, various mathematical approaches, and the applications to image and video pattern analysis and recognition. - This is a research-orientated course.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Introduction: Description of patterns, Probabilistic formulation of PR, Geometrical interpretation, Applications of PR, Scope of statistical PR. Elementary Bayes decision theory, statistical criterion and discriminant functions, linear decision functions, piecewise linear decision functions, minimum distance classifier, nonlinear classification theory, multiple pattern classification.			10
II	Representation of patterns and Feature Selection: Representation of binary random patterns – Orthogonal series expansion procedures, and Markov dependence considerations.KL expansion of patterns and its properties.Information measures of feature effectiveness, Distance measure and performance bounds, Multiclass distance measures, Feature selection criterion, Evaluation of feature subset, Algorithm of dimensionality reduction, dimensionality and sample size.			10
III	Supervised and Unsupervised Learning : Bayesian estimation for Gaussian patterns, Comments on supervised Bayesian estimation, parameter estimation of slowly varying patterns, Bayes solution to unsupervised estimation, Estimation of mixture parameters, and decision-directed estimation. Graph theoretic method, component analysis – ICA, PCA, Artificial neural network. Recursive Algorithm using stochastic approximation.			12
IV	Nonparametric methods and compound decision theory: Basic concepts and tools, Sample set construction, Nearest – Neighbor decision procedure, compound decision procedure, nonparametric estimation of multivariate density function, and nonparametric feature selection.Cluster and Mode-Seeking Techniques: Distance and similarity measures, clustering methods. Sequential pattern recognition systems: Bayes sequential decision procedure and the computational problems, Feature-ordering and selection problems, and nonparametric sequential ranking procedure. Contextual analysis in pattern recognition.			13
Texts / References Books				
- R.O. Duda, P.E. Hart and D.G. Stork, Pattern Classification, Wiley and Sons, 2001. C. H. Chen, Statistical Pattern Recognition, Spartan Books, 1973. - Christopher M. Bishop, Pattern Recognition and Machine Learning, Springer, 2016. B. Schölkopf and A.J. Smola, Learning with Kernels, MIT Press, 2002. - Jerome Friedman, Robert Tibshirani, Trevor Hastie, The Elements of Statistical Learning, Springer-Verlag , 2017.				
Learning Outcomes				
After successful completion of this course, students will be able to:				
- Understand the core theories and algorithms of statistical pattern recognition. - Understand the state-of-the-art of statistical pattern recognition. - Perform parametric classifier design, nonparametric classifier design, feature selection and dimension reduction, unsupervised data clustering. - Understand the applications such as face recognition, face detection, object detection, gesture recognition, speech recognition, etc.				

CSCMJ806	Image Analysis and Computer Vision	Theory	Practical	Cumulative
		3	1	4
Major		45	30	75
Course Objectives				
- To equip students with a comprehensive understanding of morphological image processing, including segmentation, boundary extraction, and object recognition. - To learn set theory and logic operations with binary images, and apply morphological algorithms such as dilation, erosion, opening, closing, and hit-or-miss transformation. - To explore techniques for detecting discontinuities, component labeling, and segmentation methods. - To study advanced representation techniques, including Fourier descriptors. - To emphasize real-world applications in medical image analysis, automated inspection systems, and facial recognition technology to provide practical context and enhance learning outcomes.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Morphological Image Processing: Basic concept of set theory, logic operation involving binary images, dilation and erosion, opening and closing, and hit-or-miss transformation. Some basic morphological algorithms – Boundary extraction, region filling, extraction of connected components, convex hull, thinning, thickening, skeletons, and pruning. Extensions to gray-scale images – Dilation, Erosion, Opening and closing, and application of gray scale morphology.			10
II	Image Segmentation: Detection of discontinuities – Point detection, line detection, edge detection – gradient operators, compass operators, Laplace operators and zero crossing, stochastic gradients, performance of edge detector operators. Amplitude thresholding or window slicing, component labeling, boundary based approaches, region-based approaches and clustering, template matching, and texture segmentation.			15
III	Boundary Extraction and Representation: Connectivity, Contour following, edge linking, Hough transform, chain code, fitting line segments, B-spline representation, Fourier descriptors, shape number, and autoregressive model. Region Representation: Run-length codes, quad-trees, topological descriptor, texture and projections.			10
IV	Moment Representation, Shape feature, Texture, Scene matching and matched filtering. Object recognition and image understanding: Patterns and pattern classes, decision theoretic and structural methods.			10
Texts / References Books				
- Jain, A. K. (2015). Fundamentals of digital image processing. Pearson Education India. - Gonzalez, R., & Woods, R. (2017). Digital image processing. Pearson Education India. - Forsyth, D., & Ponce, J. (2015). Computer vision: A modern approach. Pearson. - Szeliski, R. (2010). Computer vision: Algorithms and applications. Springer. - Prince, S. J. D. (2012). Computer vision: Models, learning, and inference. Cambridge University Press. - Hartley, R., & Zisserman, A. (2004). Multiple view geometry in computer vision. Cambridge University Press. - Soille, P. (2010). Morphological image analysis: Principles and applications. Springer.				
Learning Outcomes				
After successful completion of this course, students will be able to:				
- Understand and apply fundamental concepts of morphological image processing, including dilation, erosion, opening, closing, and hit-or-miss transformation. - Detect image discontinuities, perform amplitude thresholding, component labeling, and utilize various segmentation methods. - Gain proficiency in boundary extraction, edge linking, contour following, and advanced representation techniques such as Fourier descriptors and autoregressive models. - Learn to recognize objects using decision-theoretic and structural methods. - Apply these techniques to real-world scenarios like medical image analysis, automated inspection systems, and facial recognition.				

CSCMJ806	Natural Language Processing	Theory	Practical	Cumulative
		3	1	4
Major		45	30	75
Course Objectives				
- This course aims to introduce the theory and methodology of natural language understanding and generation. - Topics include stemming, lemmatization, parts of speech tagging, parsing, and machine translation. - Employing specialized libraries, students develop applications for topic modeling, sentiment analysis, and text summarization.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Introduction- Human languages, models, ambiguity, processing paradigms; Phases in natural language processing, applications, Text representation in computers, encoding schemes; Linguistics resources- Introduction to corpus, elements in balanced corpus, TreeBank, PropBank, WordNet, VerbNet etc., Resource management with XML, Management of linguistic data with the help of GATE, NLTK.			10
II	Regular expressions, Finite State Automata, word recognition, lexicon, Morphology, acquisition models, Finite State Transducer, N-grams, smoothing, entropy, HMM, ME, SVM, CRF; Part of Speech tagging- Stochastic POS tagging, HMM, Transformation based tagging (TBL), Handling of unknown words, named entities, multi word expressions.			12
III	A survey on natural language grammars, lexeme, phonemes, phrases and idioms, word order, agreement, tense, aspect and mood and agreement, Context Free Grammar, spoken language syntax; Parsing- Unification, probabilistic parsing, TreeBank, Semantics- Meaning representation, semantic analysis, lexical semantics, WordNet.			11
IV	Word Sense Disambiguation- Selectional restriction, machine learning approaches, dictionary based approaches; Discourse- Reference resolution, constraints on co-reference, algorithm for pronoun resolution, text coherence, discourse structure; Applications of NLP- Spell-checking, Summarization; Information Retrieval- Vector space model, term weighting, homonymy, polysemy, synonymy, improving user queries; Machine Translation– Overview.			12
Texts / References Books				
- Daniel Jurafsky and James H Martin. Speech and Language Processing, 2e, Pearson Education, 2009. - James A.. Natural language Understanding 2e, Pearson Education, 1994. - Bharati A., Sangal R., Chaitanya V., Natural language processing: a Paninian perspective, PHI, 2000. - Siddiqui T., Tiwary U. S., Natural language processing and Information retrieval, OUP, 2008.				
Learning Outcomes				
After successful completion of this course, students will be able to:				
- Extract information from text automatically using concepts and methods from natural language processing (NLP) including stemming, n-grams, POS tagging, and parsing. - Analyze the syntax, semantics, and pragmatics of a statement written in a natural language. - Apply machine learning algorithms to natural language processing. - Write scripts and applications in Python to carry out natural language processing using libraries such as NLTK, Gensim, and spaCY. - Design NLP-based AI systems for question answering, text summarization, and machine translation. - Evaluate the performance of NLP tools and systems.				

CSCMJ808	Data Compression	Theory	Practical	Cumulative
		3	1	4
Major		45	30	75
Course Objectives				
- To provide a comprehensive understanding of data compression techniques, including theoretical foundations and practical applications. - To explore lossless and lossy compression methods, such as Huffman coding, arithmetic coding, and quantization techniques. - To cover key image and video compression standards, including JPEG, JPEG-2000, H.264, and H.265 (HEVC). - To examine the role of machine learning in data compression, focusing on neural network-based compression and autoencoders. - To equip students with the skills to implement and evaluate various data compression algorithms.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Introduction to Data Compression: Overview of data compression, History and significance, Basic concepts and terminology. Information Theory and Entropy: Fundamentals of information theory, Entropy and information content, Source coding theorem, Applications and examples.			10
II	Lossless Compression Techniques: Huffman coding, Arithmetic coding, Run-length encoding (RLE), Lempel-Ziv-Welch (LZW) compression. Lossy Compression: Fundamentals of lossy compression, Quantization techniques, Rate-distortion theory.			10
III	Transform Coding: Discrete Cosine Transform, Wavelet transforms. Image Compression Standards: JPEG and JPEG-2000, Advanced image compression, Compression performance and quality metrics.			10
IV	Video Compression Techniques: Video coding basics, Motion estimation and compensation, Standards: H.264, H.265 (HEVC). Machine Learning in Data Compression: Neural network-based compression, Autoencoders for image compression.			15
Texts / References Books				
- Sayood, K. (2017). Introduction to Data Compression. Morgan Kaufmann. - Salomon, D., & Motta, G. (2010). Handbook of Data Compression. Springer. - Gonzalez, R. C., & Woods, R. E. (2018). Digital Image Processing. Pearson.				
Learning Outcomes				
After successful completion of this course, students will be able to: - Understand and explain the fundamental concepts and significance of data compression. - Gain the ability to apply lossless compression techniques such as Huffman coding, arithmetic coding, and LZW compression, and implement lossy compression methods, including quantization and rate-distortion theory. - Utilize transform coding techniques, including Discrete Cosine Transform and wavelet transforms, and analyze and apply image and video compression standards like JPEG, JPEG-2000, H.264, and H.265 (HEVC). - Explore the use of machine learning, particularly neural networks and autoencoders, in data compression and evaluate the performance and quality of various compression algorithms.				

CSCMJ809	Mobile Adhoc Networks	Theory	Practical	Cumulative
		3	1	4
Major		45	30	75
Course Objectives				
- To learn the core concepts, architecture, and challenges of Mobile Ad Hoc Networks, including dynamic topology, resource constraints, and key protocols. - To implement and assess routing protocols and performance metrics for MANETs, addressing issues such as network efficiency, security, and scalability.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Ad-hoc Wireless Networks: Introduction, cellular and ad-hoc wireless networks, applications of ad-hoc wireless networks, issues in ad-hoc wireless networks, ad-hoc wireless internet. MAC Protocols: Introduction, issues in designing a MAC protocol, design goals of a MAC protocol, classifications of MAC protocols, contention-based protocols, contention-based protocols with reservation mechanisms, contention-based MAC protocols with scheduling mechanisms, MAC protocols that use directional antennas.			13
II	Routing Protocols: Introduction, issues in designing a routing protocol, classifications of routing protocols, table-driven routing protocols, on-demand routing protocols, hybrid routing protocols, routing protocols with efficient flooding mechanisms, hierarchical routing protocols, power-aware routing protocols.			12
III	Multicast Routing Protocols: Introduction, issues in designing a multicast routing protocol, operation of multicast routing protocols, an architecture reference model for multicast routing protocols, classifications of multicast routing protocols, tree-based multicast routing protocols, mesh-based multicast routing protocols, energy-efficient multicasting, multicasting with quality of service guarantees, application-dependent multicast routing.			10
IV	Transport Layer and Security Protocols: Introduction, issues in designing a transport layer protocol, design goals of a transport layer protocol, classification of transport layer solutions, TCP over ad-hoc wireless networks, other transport layer protocols, security in ad-hoc wireless networks, network security requirements, issues and challenges in security provisioning, network security attacks, key management, secure routing protocols. Extension of MANET: Vehicular ad-hoc networks, Delay tolerant networks, Opportunistic networks etc.			10
Texts / References Books				
C. Siva Ram Murthy, B. S. Manoj, Ad hoc Wireless Networks: Architectures and protocols, 2nd edition, Pearson Education. 2007. C. E. Perkins, Ad hoc Networking, Addison – Wesley, 2000. - S.Basagni, M. Conti, S. Giordano, I. Stojmenovic, Mobile ad-hoc networking, Wiley-IEEE press, 2004. - M.Ilyas, The handbook of ad-hoc wireless networks, CRC press, 2002.				
Learning Outcomes				
After successful completion of this course, students will be able to: - Gain a thorough understanding of Mobile Ad Hoc Networks (MANETs), including their architecture, protocols, and key challenges such as dynamic topologies and limited resources. - Develop skills to implement and analyze various routing protocols used in MANETs, such as AODV, DSR, and OLSR. - Assess the performance of MANETs using metrics like throughput, delay, and packet delivery ratio, and apply optimization techniques to enhance network efficiency. - Identify and address security issues specific to MANETs, including encryption, authentication, and intrusion detection, to ensure network integrity and safety.				

CSCMJ810	Big Data Analytics	Theory	Practical	Cumulative
		3	1	4
		45	30	75
Major				
Course Objectives				
- This course gives an overview of Big Data, i.e. storage, retrieval and processing of big data. - In addition, it also focuses on the “technologies”, i.e., the tools/algorithms that are available for storage, processing of Big Data. - It also helps a student to perform a variety of “analytics” on different data sets and to arrive at positive conclusions.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Introduction to Big Data Analytics: Nuances of big data, Value, Issues, Case for Big data, Big data options Team challenge, Big data sources, Acquisition, Nuts and Bolts of Big data. Features of Big Data, Security, Compliance, auditing and protection, Evolution of Big data, Best Practices for Big data Analytics, Big data characteristics, Volume, Veracity, Velocity, Variety			10
II	Data Analysis: Evolution of analytics, Scalability, Analytic methods, analytic tools, Pentaho Analysis approaches, Statistical significance, Business approaches, Analytic innovation, Traditional approaches, Iterative.			10
III	Stream Computing: Introduction to Streams Concepts, Stream data model and architecture , Stream Computing, Sampling data in a stream, Filtering streams, Counting distinct elements in a stream, Estimating moments, Counting oneness in a window, Decaying window , Real-time Analytics Platform(RTAP) and applications. Data stage, Statistical analysis, Intelligent scheduler.			12
IV	Predictive Analytics: Predictive Analytics, Supervised, Unsupervised learning, Neural networks, Kohen models, Normal, Deviations from normal patterns, Normal behaviors, Expert options, Variable entry , Mining Frequent item sets, Market based model, Apriori Algorithm, Handling large data sets in Main memory, Limited Pass algorithm, Counting frequent itemsets in a stream, Clustering Techniques, Hierarchical, K-Means Clustering high dimensional data. Frameworks and Applications: IBM for Big Data, Map Reduce Framework, Hadoop, Hive, Shading, and Hadoop Distributed file systems, Hbase.			13
Texts / References Books				
- Michael Berthold, David J. Hand, Intelligent Data Analysis, Springer, 2007. - Bill Franks, “Taming the Big Data Tidal Wave: Finding Opportunities in Huge Data Streams with Advanced Analytics”, Wiley and SAS Business Series, 2012. - Paul Zikopoulos, Chris Eaton, Paul Zikopoulos, “Understanding Big Data: Analytics for Enterprise Class Hadoop and Streaming Data”, McGraw Hill, 2011. - Paul Zikopoulos, Dirk deRoos, Krishnan Parasuraman, Thomas Deutsch , James Giles, David Corrigan, “Harness the Power of Big data, The big data platform”, McGraw Hill, 2012.				
Learning Outcomes				
After successful completion of this course, students will be able to: - Understand Big Data and its analytics in the real world. - Analyze the Big Data framework like Hadoop and NOSQL to efficiently store and process Big Data to generate analytics. - Design of Algorithms to solve Data Intensive Problems using Map Reduce Paradigm. - Design and Implementation of Big Data Analytics using pig and spark to solve data intensive problems and to generate analytics. - Implement Big Data Activities using Hive and other tools.				

CSCMJ811	Quantum Computing	Theory	Practical	Cumulative
		3	1	4
Major		45	30	75
Course Objectives				
- To learn about the basics of quantum computing and its applications. - To learn the fundamentals of quantum computing and the differences between it and conventional computing. - To understand the principals of Quantum Computing.				
Syllabus				
Units	Course Content			Hr. of Teaching
I	Introduction to Quantum Computing: Introduction of computing and Quantum Computing-Need of Quantum Computing-Deutsch’s algorithm.			08
II	Linear Operators and Spectral Decomposition: Operators: Matrices as linear operators, Examples of matrices, Kernel, image and rank. Operations on matrices: Trace, Entry-wise multiplication, Inverse, Eigen values and eigen-vectors. Spectralde composition: Hermitian matrix, Unitary matrix, Positive semi-definite matrix, Singular value decomposition, Simultaneously diagonalizable matrices.			15
III	Operator Functions, Tensor Products and Postulates of Quantum mechanics: Operator functions, Tensor product, Direct sum, State of a system, Evolution of the system, Measurement of the system, Composite Systems, Quantum Teleportation.			10
IV	Computation, Quantum and Classical: Classical circuits-Quantum circuits–Algorithms: Randomized algorithms, Approximation algorithms. Deutsch-Josza, Fourier Transform and Phase Estimation: Deutsch-Jozsaalgorithm: Bernstein-Vazirani algorithm- Fourier transform: Discrete Fourier transform, Fast Fourier transform- Quantum discrete Fourier transform- Phase estimation.			12
Texts / References Books				
1. Nielsen, M. A., & Chuang, I. L. (2019).Quantum computation and quantum information. Cambridge University Press. 2. Kaye, P., Laflamme, R., & Mosca, M. (2020). An introduction to quantum computing. Oxford University Press.				
Learning Outcomes				
On completion of this course, the student will be able to: - Use the mathematical framework of quantum computing to solve computational problems. - Critically read and understand scientific papers on quantum computing. - Explain and analyze any quantum algorithms described in quantum circuit or measurement-based quantum computing models. - Relate quantum complexity classes to the classical ones. - Gain experience in problem solving for complex system.				